

Performance Analysis

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Detailed analysis of your performance in the AWS Certified Solutions Architect Associate - SAA C03 exam.

 Answer Distribution

0

Correct

1

Incorrect

64

Unattempted

 Domain Performance

Storage	0%
Monitoring and Logging	0%
Security	0%
Networking	0%
Serverless	0%
Databases	0%
Compute	0%
Disaster Recovery	0%

Final Score

**0/65**

Passing score: 48

Time Taken

**130m 0s**

Total time: 130 minutes

Question Analysis

✓ Correct	0
✗ Incorrect	1
⊖ Unattempted	64
🔖 Reviewed	0

Question-wise Analysis

Expand All

Collapse All

Question 1

Storage



A telemarketing company is building its customer call center functionality on AWS. The company requires a solution that offers multiple speaker recognition and generates transcript files. The company intends to query these transcript files to analyze business patterns. The transcript files must be stored for 7 years to meet auditing requirements. Which solution satisfies these requirements?

- ⊗ Use Amazon Rekognition for multiple speaker recognition. Store the transcript files in Amazon S3. Use machine learning models for transcript file analysis.

Amazon Rekognition is designed for image and video analysis, not for transcribing audio or recognizing multiple speakers.

- ✔ Use Amazon Transcribe for multiple speaker recognition. Store the transcript files in Amazon S3. Use Amazon Athena for transcript file analysis.

Amazon Transcribe can handle multiple speakers and generate transcript files, which can be stored in Amazon S3. Amazon Athena can then be used to query the transcript files in S3 for business pattern analysis. The 7 year retention can be configured on S3.

- ⊗ Use Amazon Translate for multiple speaker recognition. Store the transcript files in Amazon Redshift. Use SQL queries for transcript file analysis.

Amazon Translate is a service for translating text from one language to another, not transcribing spoken language. Amazon Redshift is also an overkill for storing just transcript files.

- ⊗ Use Amazon Rekognition for multiple speaker recognition. Store the transcript files in Amazon S3. Use Amazon Textract for transcript file analysis.

Amazon Rekognition is for image and video analysis. Amazon Textract extracts text and data from documents and images, but not for transcribing speech.

Overall Explanation:

Amazon Transcribe is the service best suited for transcribing spoken language into written text, supporting multiple speakers. Amazon Athena is a query service for analyzing data in Amazon S3 using SQL, ideal for analyzing transcript files. Storing the files in S3 also allows you to meet the 7 year auditing requirements. Therefore, the correct answer is B.

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Question 2

Not Attempted

Monitoring and Logging



A company needs the ability to analyze the log files of its proprietary application. The logs are stored in JSON format in an Amazon S3 bucket. Queries will be simple and will run on-demand. A solutions architect needs to perform the analysis with minimal changes to the existing architecture and operational overhead. What should the solutions architect do to meet these requirements?

- ⊗ Use Amazon Redshift to load all the content into one place and run the SQL queries as needed.

This option involves an extra step of loading the data into Redshift, which is unnecessary for simple queries and increases operational overhead.

- ⊗ Use Amazon CloudWatch Logs to store the logs. Run SQL queries as needed from the Amazon CloudWatch console.

CloudWatch Logs is primarily for log monitoring and management, not ad-hoc querying of JSON data. It is not the most efficient solution for the given requirements.

- ✓ Use Amazon Athena directly with Amazon S3 to run the queries as needed.

Amazon Athena allows querying data directly in S3 using SQL, making it ideal for on-demand analysis of JSON logs with minimal setup and operational overhead.

- ⊗ Use AWS Glue to catalog the logs. Use a transient Apache Spark cluster on Amazon EMR to run the SQL queries as needed.

This option involves setting up and managing an EMR cluster, which adds significant operational overhead and is not necessary for simple queries.

Overall Explanation:

Amazon Athena is the most suitable solution because it directly queries the JSON logs stored in S3 using standard SQL, requiring minimal configuration and operational overhead. It's serverless and optimized for on-demand queries, fitting the described scenario perfectly. Redshift, CloudWatch Logs, and EMR all introduce unnecessary complexity and overhead.

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Question 3

Not Attempted

Storage



A company needs to clone a large production dataset into a test environment within the same AWS Region. The data resides on Amazon EC2 instances using Amazon Elastic Block Store (Amazon EBS) volumes. Changes made to the cloned data should not impact the production environment. The application using this data demands consistently high I/O performance. The solutions architect must minimize the time required to clone the production data into the test environment. Which solution meets these requirements most effectively?

- ⊗ Take EBS snapshots of the production EBS volumes. Restore the snapshots onto EC2 instance store volumes in the test environment.

Restoring to instance store volumes might offer high I/O, but instance store volumes are ephemeral and not suitable for persistent data storage and may not be as durable as EBS volumes.

- ⊗ Configure the production EBS volumes to use the EBS Multi-Attach feature. Take EBS snapshots of the production EBS volumes. Attach the production EBS volumes to the EC2 instances in the test environment.

Using EBS Multi-Attach with production volumes directly in the test environment would violate the requirement that modifications in the test environment must not affect the production environment.

- ⊗ Take EBS snapshots of the production EBS volumes. Create and initialize new EBS volumes. Attach the new EBS volumes to EC2 instances in the test environment before restoring the volumes from the production EBS snapshots.

Creating and attaching volumes before restoring the snapshot is not a standard or efficient procedure. The restore process creates the new volume. Also, the data will not be ready to use as soon as the restore process completes.

- ✔ Take EBS snapshots of the production EBS volumes. Turn on the EBS fast snapshot restore feature on the EBS snapshots. Restore the snapshots into new EBS volumes. Attach the new EBS volumes to EC2 instances in the test environment.

This option uses EBS fast snapshot restore to minimize the time to create fully initialized EBS volumes from snapshots, providing consistent high I/O performance immediately after creation.

Overall Explanation:

The most efficient solution is to utilize EBS snapshots in conjunction with the Fast Snapshot Restore (FSR) feature. Taking snapshots creates a point-in-time copy of the production data. Enabling FSR ensures that new volumes created from these snapshots are fully initialized upon creation, eliminating initial latency and delivering consistently high I/O performance immediately. The newly created EBS volumes are then attached to the test environment EC2 instances, isolating the production data from any modifications made during testing.

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Question 4

Not Attempted

Security



A company hosts its application on AWS. The company uses Amazon Cognito to manage users. When users log in to the application, the application fetches required data from Amazon DynamoDB by using a REST API that is hosted in Amazon API Gateway. The company wants an AWS managed solution that will control access to the REST API to reduce development efforts. Which solution will meet these requirements with the LEAST operational overhead?

- ☐ **Configure an AWS Lambda function to be an authorizer in API Gateway to validate which user made the request.**

Using a Lambda authorizer requires developing and maintaining custom code, increasing operational overhead.

- ☐ **For each user, create and assign an API key that must be sent with each request. Validate the key by using an AWS Lambda function.**

Managing API keys for each user adds significant operational complexity and is not the most efficient solution.

- ☐ **Send the user's email address in the header with every request. Invoke an AWS Lambda function to validate that the user with that email address has proper access.**

Sending the user's email in the header and validating it with a Lambda function is not a secure and efficient practice, and adds unnecessary overhead.

- ☒ **Configure an Amazon Cognito user pool authorizer in API Gateway to allow Amazon Cognito to validate each request.**

Integrating API Gateway with a Cognito User Pool authorizer provides a managed and streamlined solution for access control with minimal operational overhead. API Gateway handles the validation of tokens from Cognito.

Overall Explanation:

The best solution is to configure an Amazon Cognito user pool authorizer in API Gateway. This allows Amazon Cognito to handle the user validation, leveraging the existing user management infrastructure and minimizing development and operational overhead. API Gateway handles the validation automatically.

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Question 5

Not Attempted

Networking



A company is developing a marketing communications service that targets mobile app users. The company needs to send confirmation messages with Short Message Service (SMS) to its users. The users must be able to reply to the SMS messages. The company must store the responses for a year for analysis. What should a solutions architect do to meet these requirements?

- ☐ Create an Amazon Connect contact flow to send the SMS messages. Use AWS Lambda to process the responses.

Amazon Connect is primarily designed for customer support and engagement, not for storing and processing SMS responses for later analysis.

- ☒ Build an Amazon Pinpoint journey. Configure Amazon Pinpoint to send events to an Amazon Kinesis data stream for analysis and archiving.

Amazon Pinpoint allows sending SMS messages, receiving replies, and streaming events to Kinesis for analysis and archiving, meeting all requirements.

- ☐ Use Amazon Simple Queue Service (Amazon SQS) to distribute the SMS messages. Use AWS Lambda to process the responses.

SQS is a message queuing service and not suited for SMS communication or capturing responses for analysis.

- ☐ Create an Amazon Simple Notification Service (Amazon SNS) FIFO topic. Subscribe an Amazon Kinesis data stream to the SNS topic for analysis and archiving.

SNS is primarily for message publishing and distribution and is not the best choice for managing two-way SMS communication and storing responses.

Overall Explanation:

Amazon Pinpoint is designed for marketing communications and supports sending SMS messages, receiving replies, and integrating with Amazon Kinesis for storing and analyzing the responses. Therefore, building an Amazon Pinpoint journey and configuring it to send events to Kinesis is the most appropriate solution.

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Question 6

Not Attempted

Storage



A company is planning to move its data to an Amazon S3 bucket. The data must be encrypted at rest. Additionally, the encryption key must be automatically rotated every year, with the LEAST operational overhead.

- ✔ Move the data to the S3 bucket. Use server-side encryption with Amazon S3 managed encryption keys (SSE-S3). Use the built-in key rotation behavior of SSE-S3 encryption keys.

SSE-S3 uses keys managed by Amazon S3, with automatic key rotation approximately every year. This option provides the least operational overhead as it requires no additional configuration or maintenance for key rotation.

- ✘ Create an AWS Key Management Service (AWS KMS) customer managed key. Enable automatic key rotation. Set the S3 bucket's default encryption behavior to use the customer managed KMS key. Move the data to the S3 bucket.

While this meets the encryption and rotation requirements, using a customer-managed KMS key involves more operational overhead than SSE-S3, including costs associated with KMS and management of the key itself.

- ✘ Create an AWS Key Management Service (AWS KMS) customer managed key. Set the S3 bucket's default encryption behavior to use the customer managed KMS key. Move the data to the S3 bucket. Manually rotate the KMS key every year.

Manually rotating the KMS key adds significant operational overhead and introduces the risk of human error.

- ✘ Encrypt the data with customer key material before moving the data to the S3 bucket. Create an AWS Key Management Service (AWS KMS) key without key material. Import the customer key material into the KMS key. Enable automatic key rotation.

This solution involves encrypting the data prior to upload and importing key material, increasing complexity and operational overhead compared to SSE-S3 or KMS with AWS-managed keys.

Overall Explanation:

The best solution with the LEAST operational overhead is to use server-side encryption with Amazon S3 managed encryption keys (SSE-S3). SSE-S3 handles key management and automatic rotation approximately every year, requiring minimal configuration. Using customer managed keys, even with automatic rotation enabled, involves more overhead in terms of cost and key management. The other options involve manual intervention or pre-encryption, which adds unnecessary complexity.

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Question 7

Not Attempted

Serverless



A finance company uses a web application on Amazon EC2 instances to accept customer appointment requests via text messages. These requests are published to an Amazon SQS queue. Another application on EC2 instances processes these requests, sending meeting invitations and confirmation emails, and then stores the meeting information in Amazon DynamoDB. Recently, customers have reported delays in receiving their meeting invitations as the company has scaled. What is the MOST effective solution a solutions architect should recommend to address this issue?

- ☐ Add a DynamoDB Accelerator (DAX) cluster in front of the DynamoDB database.
Adding DAX improves DynamoDB read performance but doesn't directly address the bottleneck in processing SQS messages and sending invitations.
- ☐ Add an Amazon API Gateway API in front of the web application that accepts the appointment requests.
API Gateway manages incoming requests to the web application, but it doesn't directly address the processing of messages in the SQS queue or the sending of invitations.
- ☐ Add an Amazon CloudFront distribution. Set the origin as the web application that accepts the appointment requests.
CloudFront improves content delivery and caching for the web application, but doesn't solve the delay in processing appointment requests and sending meeting invitations.
- ☒ Add an Auto Scaling group for the application that sends meeting invitations.
Configure the Auto Scaling group to scale based on the depth of the SQS queue.
Adding an Auto Scaling group for the invitation-sending application, scaling based on SQS queue depth, allows the system to automatically adjust capacity to handle increased message load and process invitations more efficiently, reducing delays.

Overall Explanation:

The bottleneck lies in the application processing the SQS queue and sending meeting invitations. By adding an Auto Scaling group that scales based on the SQS queue depth, the system can dynamically increase its capacity to handle more requests concurrently,

reducing the delay in sending meeting invitations. Options A, B, and C do not directly address this processing bottleneck.

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Question 8

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Databases



An online retail company has more than 50 million active customers and receives more than 25,000 orders each day. The company collects purchase data for customers and stores this data in Amazon S3. Additional customer data is stored in Amazon RDS. The company wants to make all the data available to various teams so that the teams can perform analytics. The solution must provide the ability to manage fine-grained permissions for the data and must minimize operational overhead. Which solution will meet these requirements?

- ☐ Migrate the purchase data to write directly to Amazon RDS. Use RDS access controls to limit access.

Writing purchase data directly to Amazon RDS with RDS access controls lacks comprehensive permissions management for both S3 and RDS data and may impact the RDS performance.

- ☐ Schedule an AWS Lambda function to periodically copy data from Amazon RDS to Amazon S3. Create an AWS Glue crawler. Use Amazon Athena to query the data. Use S3 policies to limit access.

Periodically copying data from RDS to S3 using Lambda and using AWS Glue and Athena for querying does not offer fine-grained permissions management and introduces data synchronization complexities.

- ☒ Create a data lake by using AWS Lake Formation. Create an AWS Glue JDBC connection to Amazon RDS. Register the S3 bucket in Lake Formation. Use Lake Formation access controls to limit access.

Lake Formation enables the creation of a secure and scalable data lake on AWS, allowing centralized access controls for both S3 and RDS data. By using Lake Formation, the company can manage permissions effectively and integrate RDS data through the AWS Glue JDBC connection. Registering the S3 in Lake Formation ensures unified access control. This solution reduces operational overhead while providing fine-grained permissions management.

- ☐ Create an Amazon Redshift cluster. Schedule an AWS Lambda function to periodically copy data from Amazon S3 and Amazon RDS to Amazon Redshift. Use Amazon Redshift access controls to limit access.

Creating an Amazon Redshift cluster and copying data from S3 and RDS to Redshift adds complexity and operational overhead without the flexibility of Lake Formation's permissions management capabilities.

Overall Explanation:

AWS Lake Formation allows creating a secure and scalable data lake with centralized, fine-grained access control for both S3 and RDS data. It minimizes operational overhead while providing the required permissions management capabilities. The key is the 'manage fine-grained permissions for the data' requirement, which Lake Formation addresses directly.

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Question 9

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Storage



A company hosts a marketing website in an on-premises data center. The website consists of static documents and runs on a single server. An administrator updates the website content infrequently and uses an SFTP client to upload new documents. The company decides to host its website on AWS and to use Amazon CloudFront. The solutions architect must design the most cost-effective and resilient architecture for website hosting to serve as the CloudFront origin. The solutions architect also needs to ensure secure access to the website content. Which solution will meet these requirements?

- ☐ Create a virtual server by using Amazon Lightsail. Configure the web server in the Lightsail instance. Upload website content by using an SFTP client.
Lightsail introduces unnecessary overhead and costs compared to using S3 directly for static content.
- ☐ Create an AWS Auto Scaling group for Amazon EC2 instances. Use an Application Load Balancer. Upload website content by using an SFTP client.
Using an Auto Scaling group with EC2 instances and an ALB is overkill for serving static content and adds unnecessary complexity and cost.
- ☒ Create a private Amazon S3 bucket. Use an S3 bucket policy to allow access from a CloudFront origin access identity (OAI). Upload website content by using the AWS CLI.
Hosting the website in a private S3 bucket provides cost-effective and highly available storage for static content. Configuring a bucket policy for OAI ensures secure access only through CloudFront.

- ⊗ Create a public Amazon S3 bucket. Configure AWS Transfer for SFTP. Configure the S3 bucket for website hosting. Upload website content by using the SFTP client.

Using a public S3 bucket and AWS Transfer for SFTP introduces additional costs and complexity and is less secure than using a private bucket with an OAI.

Overall Explanation:

The most cost-effective and resilient solution is to use a private S3 bucket with a CloudFront origin access identity (OAI). This allows secure access to the static website content via CloudFront and utilizes the AWS CLI for content uploads. Option A and B involve more complex and expensive resources. Option D suggests a public S3 bucket and AWS Transfer which isn't as secure and adds complexity and cost.

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Question 10

Not Attempted

Monitoring and Logging



A company wants to manage Amazon Machine Images (AMIs). The company currently copies AMIs to the same AWS Region where the AMIs were created. The company needs to design an application that captures AWS API calls and sends alerts whenever the Amazon EC2 `CreateImage` API operation is called within the company's account. Which solution will meet these requirements with the LEAST operational overhead?

- ⊗ Create an AWS Lambda function to query AWS CloudTrail logs and to send an alert when a `CreateImage` API call is detected.
- Creating a Lambda function to query CloudTrail adds operational overhead for management and maintenance of the Lambda function.
- ⊗ Configure AWS CloudTrail with an Amazon Simple Notification Service (Amazon SNS) notification that occurs when updated logs are sent to Amazon S3. Use Amazon Athena to create a new table and to query on `CreateImage` when an API call is detected.
- This option involves using CloudTrail, S3, SNS and Athena, adding unnecessary overhead.
- ✔ Create an Amazon EventBridge (Amazon CloudWatch Events) rule for the `CreateImage` API call. Configure the target as an Amazon Simple Notification Service (Amazon SNS) topic to send an alert when a `CreateImage` API call is detected.

EventBridge directly integrates with CloudTrail events and can trigger an SNS topic with minimal configuration and overhead.

- ⊗ Configure an Amazon Simple Queue Service (Amazon SQS) FIFO queue as a target for AWS CloudTrail logs. Create an AWS Lambda function to send an alert to an Amazon Simple Notification Service (Amazon SNS) topic when a `CreateImage` API call is detected.

This option uses SQS, Lambda, and SNS, increasing the complexity and operational overhead of the solution. EventBridge is a more direct solution.

Overall Explanation:

The most efficient solution with the least operational overhead is to use Amazon EventBridge. EventBridge (formerly CloudWatch Events) can directly capture API calls using CloudTrail events and trigger an SNS topic. This avoids the need for Lambda functions, Athena queries, or SQS queues, simplifying the setup and reducing management overhead. Option C is the right answer.

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Question 11

Not Attempted

Databases



A company has an asynchronous API deployed using Amazon API Gateway, invoking an AWS Lambda function that stores user requests in Amazon DynamoDB before dispatching them to microservices. Despite provisioning DynamoDB throughput within budget, the company is still losing user requests due to availability issues. How can a solutions architect address this problem without negatively affecting current users?

- ⊗ Add throttling to the API Gateway with server-side throttling limits.
Throttling may lead to increased errors and negatively impact the user experience. It may not sufficiently address the availability issues and prevent the loss of requests.
- ⊗ Use DynamoDB Accelerator (DAX) and Lambda to buffer writes to DynamoDB.
DAX primarily improves read performance and does not directly address the issue of lost writes to DynamoDB. DAX is a read-through/write-through cache.
- ⊗ Create a secondary index in DynamoDB for the table with the user requests.

A secondary index can improve query performance but does not prevent the loss of requests due to write capacity limitations.

- ✔ Use Amazon Simple Queue Service (Amazon SQS) and Lambda to buffer writes to DynamoDB.

SQS provides a buffer for incoming requests, decoupling the API from DynamoDB write operations. Lambda can then process messages from SQS at a controlled rate, preventing DynamoDB from being overwhelmed. This prevents data loss and increases system resilience.

Overall Explanation:

Using SQS and Lambda to buffer writes decouples the API from the DynamoDB writes, and allows Lambda to manage the writes to DynamoDB at a controlled rate. This prevents data loss due to DynamoDB throttling. The asynchronous nature of the API makes this solution feasible and minimally invasive for existing users.

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Question 12

Not Attempted

Networking



A company needs to move data from an Amazon EC2 instance to an Amazon S3 bucket. The company must ensure that no API calls and no data are routed through public internet routes. Only the EC2 instance can have access to upload data to the S3 bucket. Which solution will meet these requirements?

- ✔ Create an interface VPC endpoint for Amazon S3 in the subnet where the EC2 instance is located. Attach a resource policy to the S3 bucket to only allow the EC2 instance's IAM role for access.

This option involves creating an interface endpoint for S3. Interface endpoints use private IP addresses and allow secure access to S3 without traversing the public internet. By attaching a resource policy to the S3 bucket that restricts access to the EC2 instance's IAM role, you ensure that only the EC2 instance can upload data.

- ✘ Create a gateway VPC endpoint for Amazon S3 in the Availability Zone where the EC2 instance is located. Attach appropriate security groups to the endpoint. Attach a resource policy to the S3 bucket to only allow the EC2 instance's IAM role for access.

While gateway endpoints provide private connectivity to S3, they are created at the VPC level, not at the Availability Zone level. Also, you cannot attach security groups to gateway VPC endpoints.

- ⊗ Run the nslookup tool from inside the EC2 instance to obtain the private IP address of the S3 bucket's service API endpoint. Create a route in the VPC route table to provide the EC2 instance with access to the S3 bucket. Attach a resource policy to the S3 bucket to only allow the EC2 instance's IAM role for access.

This option is not a recommended approach. S3's IP addresses can change, and relying on specific IP addresses is not a scalable or reliable solution. VPC Endpoints are the preferred method for private S3 access.

- ⊗ Use the AWS provided, publicly available ip-ranges.json file to obtain the private IP address of the S3 bucket's service API endpoint. Create a route in the VPC route table to provide the EC2 instance with access to the S3 bucket. Attach a resource policy to the S3 bucket to only allow the EC2 instance's IAM role for access.

Similar to option C, this option is also not a recommended approach. While the ip-ranges.json file provides IP address ranges for AWS services, relying on them is not a reliable solution because IPs can change. VPC Endpoints are the recommended approach.

Overall Explanation:

The correct answer is to create an interface VPC endpoint for Amazon S3. This ensures that traffic to S3 does not traverse the public internet and that access is restricted to the EC2 instance through the S3 bucket's resource policy. Gateway endpoints do not allow for security groups to be attached and interface endpoints are designed for private connectivity to AWS services like S3.

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Question 13

Not Attempted

Databases



A solutions architect is designing the architecture of a new application being deployed to the AWS Cloud. The application will run on Amazon EC2 On-Demand Instances and will automatically scale across multiple Availability Zones. The EC2 instances will scale up and down frequently throughout the day. An Application Load Balancer (ALB) will handle the load distribution. The architecture needs to support distributed session data management. The company is willing to make changes to code if needed. What should the solutions architect do to ensure that the architecture supports distributed session data management?

- ✔ Use Amazon ElastiCache to manage and store session data.

Amazon ElastiCache is a managed in-memory data store service well-suited for managing session data in a distributed architecture. It allows multiple EC2 instances to access and share session data seamlessly, providing high-performance, scalable, and durable storage. Since the company is willing to make code changes, this solution is viable.

⊗ Use session affinity (sticky sessions) of the ALB to manage session data.

Session affinity (sticky sessions) of the ALB ties each session to a specific EC2 instance. As the instances scale up and down frequently, it can lead to uneven load distribution and may not provide optimal scalability. If an instance goes down, the session data is lost.

⊗ Use Session Manager from AWS Systems Manager to manage the session.

Session Manager from AWS Systems Manager is primarily used for secure remote shell access to EC2 instances and is not designed for managing application session data.

⊗ Use the GetSessionToken API operation in AWS Security Token Service (AWS STS) to manage the session.

The GetSessionToken API operation in AWS STS is used for temporary security credentials and not session data management.

Overall Explanation:

Amazon ElastiCache is the best option for managing distributed session data in this scenario. It provides a scalable and highly available solution that allows the application to automatically scale across multiple Availability Zones without losing session data. Session affinity is not ideal due to the frequent scaling of EC2 instances. Session Manager and GetSessionToken are not applicable for session data management.

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Question 14

Not Attempted

Serverless



An e-commerce company plans to launch a one-deal-a-day website on AWS. Each day, exactly one product will be on sale for 24 hours. The company anticipates millions of requests per hour during peak hours and requires millisecond latency. Which solution meets these requirements with the LEAST operational overhead?

⊗ Use Amazon S3 to host the full website in different S3 buckets. Add Amazon CloudFront distributions. Set the S3 buckets as origins for the distributions. Store the order data in Amazon S3.

Storing order data directly in S3 is not suitable for dynamic data and transactional processing. Hosting a full website in different S3 buckets increases complexity without significant benefit.

- ⊗ Deploy the full website on Amazon EC2 instances that run in Auto Scaling groups across multiple Availability Zones. Add an Application Load Balancer (ALB) to distribute the website traffic. Add another ALB for the backend APIs. Store the data in Amazon RDS for MySQL.

This solution involves more operational overhead due to managing EC2 instances, Auto Scaling groups, and RDS for MySQL. Scaling EC2 instances and RDS can also introduce latency.

- ⊗ Migrate the full application to run in containers. Host the containers on Amazon Elastic Kubernetes Service (Amazon EKS). Use the Kubernetes Cluster Autoscaler to increase and decrease the number of pods to process bursts in traffic. Store the data in Amazon RDS for MySQL.

While containerization offers flexibility, managing an EKS cluster adds significant operational overhead. RDS for MySQL also has limitations in scaling speed for sudden bursts.

- ✓ Use an Amazon S3 bucket to host the website's static content. Deploy an Amazon CloudFront distribution. Set the S3 bucket as the origin. Use Amazon API Gateway and AWS Lambda functions for the backend APIs. Store the data in Amazon DynamoDB.

This solution utilizes serverless and managed services, minimizing operational overhead. S3 and CloudFront handle static content efficiently, API Gateway and Lambda provide scalable backend APIs, and DynamoDB offers fast, scalable NoSQL data storage.

Overall Explanation:

The optimal solution is to leverage AWS managed services: S3 for static content, CloudFront for distribution, API Gateway and Lambda for backend APIs, and DynamoDB for data storage. These services provide scalability, low latency, and require minimal operational overhead. Options involving EC2, RDS, or EKS require more manual management and scaling efforts.

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Question 15

Not Attempted

[Compute](#)

A company's food delivery service is experiencing scaling issues during peak hours. The current architecture includes EC2 instances in Auto Scaling groups for collecting and fulfilling orders. Order collection is fast, but fulfillment is slow, leading to backlogs. Data loss during scaling

events is unacceptable. A solutions architect must ensure proper scaling for both processes while optimizing resource utilization. Which solution meets these requirements?

- ⊗ Use Amazon CloudWatch metrics to monitor the CPU of each instance in the Auto Scaling groups. Configure each Auto Scaling group's minimum capacity according to peak workload values.

This solution does not dynamically scale based on actual queue backlog, potentially leading to over-provisioning or under-provisioning. It also relies on CPU utilization, which may not accurately reflect order fulfillment backlog.

- ⊗ Use Amazon CloudWatch metrics to monitor the CPU of each instance in the Auto Scaling groups. Configure a CloudWatch alarm to invoke an Amazon Simple Notification Service (Amazon SNS) topic that creates additional Auto Scaling groups on demand.

Similar to option A, this solution relies on CPU utilization and does not directly address the queue backlog. The SNS topic and on-demand scaling of Auto Scaling groups might introduce delays.

- ⊗ Provision two Amazon Simple Queue Service (Amazon SQS) queues: one for order collection and another for order fulfillment. Configure the EC2 instances to poll their respective queue. Scale the Auto Scaling groups based on notifications that the queues send.

SQS does not send notifications. EC2 instances need to poll for messages. Scaling based on the number of 'notifications' does not make logical sense. Both collection and fulfillment instances would scale in parallel even though fulfillment is the bottleneck. We should scale when a backlog is building, not just based on incoming requests.

- ✔ Provision two Amazon Simple Queue Service (Amazon SQS) queues: one for order collection and another for order fulfillment. Configure the EC2 instances to poll their respective queue. Create a metric based on a backlog per instance calculation. Scale the Auto Scaling groups based on this metric.

This solution utilizes SQS for decoupling and reliable queuing. Scaling based on the backlog per instance allows for dynamic scaling that is proportional to the load on each instance. This optimizes resource utilization and prevents data loss.

Overall Explanation:

The correct answer is D. Provisioning SQS queues decouples the system and allows for reliable queuing of orders. Scaling the Auto Scaling groups based on the backlog per instance ensures that resources are dynamically scaled based on the workload. This approach optimizes resource utilization and prevents data loss. Using SQS queues for

order collection and fulfillment, creating a metric based on backlog per instance calculation, and scaling the Auto Scaling groups accordingly, is the most suitable solution to address the scaling problems while optimizing resource utilization and ensuring reliable message processing. When the backlog per instance reaches the target value, a scale-out event will happen. Because the backlog per instance is already 150 messages (1500 messages / 10 instances), your group scales out, and it scales out by five instances to maintain proportion to the target value.

<https://docs.aws.amazon.com/autoscaling/ec2/userguide/as-using-sqs-queue.html>

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Question 16

Not Attempted

Monitoring and Logging



A company hosts multiple production applications. One application utilizes Amazon EC2, AWS Lambda, Amazon RDS, Amazon SNS, and Amazon SQS across multiple AWS Regions. All company resources are tagged with "application" as the tag name and a value corresponding to each application. A solutions architect needs to quickly identify all tagged components. Which solution provides the fastest method for achieving this?

- ☐  Use AWS CloudTrail to generate a list of resources with the application tag.
CloudTrail primarily focuses on capturing API activity. It doesn't offer a comprehensive and quick method to identify all tagged components across multiple services and Regions.
- ☐  Use the AWS CLI to query each service across all Regions to report the tagged components.
Manually querying each service using the AWS CLI is time-consuming and cumbersome, especially across multiple services and Regions.
- ☐  Run a query in Amazon CloudWatch Logs Insights to report on the components with the application tag.
CloudWatch Logs Insights focuses on analyzing logs, not directly identifying tagged components across different services and Regions.
- ☒  Run a query with the AWS Resource Groups Tag Editor to report on the resources globally with the application tag.
The Resource Groups Tag Editor is specifically designed for managing and organizing resources based on tags, providing a centralized and efficient way to generate a report of tagged components

across multiple services and Regions.

Overall Explanation:

The quickest solution is to use the AWS Resource Groups Tag Editor. This service is designed for managing and organizing resources using tags, enabling a fast and efficient way to identify all components tagged with 'application' across multiple services and regions. CloudTrail is for API activity, AWS CLI requires manual querying, and CloudWatch Logs Insights focuses on log analysis, making them less efficient for this task.

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Question 17

Not Attempted

Storage



A company needs to export its database once a day to Amazon S3 for other teams to access. The exported object size varies between 2 GB and 5 GB. The S3 access pattern for the data is variable and changes rapidly. The data must be immediately available and must remain accessible for up to 3 months. The company needs the most cost-effective solution that will not increase retrieval time. Which S3 storage class should the company use to meet these requirements?



S3 Intelligent-Tiering

S3 Intelligent-Tiering automatically moves data to the most cost-effective tier based on access patterns, ensuring immediate availability and no retrieval fees while adapting to variable access.



S3 Glacier Instant Retrieval

S3 Glacier Instant Retrieval is suitable for long-lived, rarely accessed data, but might not be the most cost-effective when the access pattern is variable and speed is crucial.



S3 Standard

S3 Standard offers immediate availability and handles variable access patterns well but might not be the most cost-effective for data that could potentially be accessed less frequently over time.



S3 Standard-Infrequent Access (S3 Standard-IA)

S3 Standard-IA is more cost-effective for infrequently accessed data but incurs retrieval fees, potentially increasing costs if the access pattern is variable and changes rapidly.

Overall Explanation:

S3 Intelligent-Tiering is the best choice because it automatically transitions data between different access tiers based on usage patterns. This ensures immediate availability and cost-effectiveness by moving infrequently accessed data to lower-cost tiers, while frequently accessed data remains in the standard tier. This aligns with rapidly changing access patterns, which helps the organization meet the requirements of cost optimization.

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Question 18

Not Attempted

Security



A company is developing a new mobile app and wants to protect its Application Load Balancer (ALB) from common application-level attacks such as cross-site scripting (XSS) and SQL injection. The company has minimal infrastructure and operational staff and wants to minimize the management overhead of securing its AWS environment. Which solution would BEST meet these requirements?

- ☒ **Configure AWS WAF rules and associate them with the ALB.**
AWS WAF provides a managed service to filter and block malicious traffic before it reaches the application, protecting against XSS and SQL injection attacks. It reduces the management overhead by eliminating the need to manage servers.
- ☐ **Deploy the application using Amazon S3 with public hosting enabled.**
Amazon S3 with public hosting is suitable for static content but does not provide the necessary security and traffic filtering capabilities to protect against application-level attacks.
- ☐ **Deploy AWS Shield Advanced and add the ALB as a protected resource.**
AWS Shield Advanced is primarily focused on DDoS protection and, while it includes WAF, it doesn't directly address the application-level attack requirements as efficiently as a targeted WAF configuration. Also, Shield Advanced is significantly more expensive than WAF.
- ☐ **Create a new ALB that directs traffic to an Amazon EC2 instance running a third-party firewall, which then passes the traffic to the current ALB.**
This option increases management overhead by requiring the company to manage and secure an EC2 instance and a third-party firewall, which contradicts the requirement of minimizing operational responsibilities.

Overall Explanation:

Configuring AWS WAF rules and associating them with the ALB is the BEST solution. AWS WAF directly addresses the application-level attacks (XSS, SQL injection) and is a managed service, reducing operational overhead. While AWS Shield Advanced also includes WAF, it is primarily for DDoS protection and more expensive. S3 hosting does not protect against application attacks, and using a third-party firewall on EC2 increases operational burden.

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Question 19

Not Attempted

Serverless



A company's reporting system delivers hundreds of .csv files to an Amazon S3 bucket each day. The company must convert these files to Apache Parquet format and must store the files in a transformed data bucket. Which solution will meet these requirements with the LEAST development effort?

- ☐ Create an Amazon EMR cluster with Apache Spark installed. Write a Spark application to transform the data. Use EMR File System (EMRFS) to write files to the transformed data bucket.

This option requires significant development effort to write and maintain a Spark application, as well as manage the EMR cluster. EMRFS adds complexity.

- ☒ Create an AWS Glue crawler to discover the data. Create an AWS Glue extract, transform, and load (ETL) job to transform the data. Specify the transformed data bucket in the output step.

AWS Glue provides a fully managed ETL service and allows for easy CSV to Parquet conversion with minimal development. Crawlers automatically discover the schema, and ETL jobs can be configured visually or with code.

- ☐ Use AWS Batch to create a job definition with Bash syntax to transform the data and output the data to the transformed data bucket. Use the job definition to submit a job. Specify an array job as the job type.

This option requires more development effort in writing and managing custom Bash scripts for data transformation and setting up AWS Batch infrastructure. It lacks the built-in capabilities of AWS Glue for data transformation.

- ⊗ Create an AWS Lambda function to transform the data and output the data to the transformed data bucket. Configure an event notification for the S3 bucket. Specify the Lambda function as the destination for the event notification.

While Lambda can transform the data, it requires more development effort to write and manage the transformation code compared to AWS Glue, which is specifically designed for ETL operations. Lambda functions also have execution time limits.

Overall Explanation:

AWS Glue is the best option because it is a fully managed ETL service that simplifies the process of preparing and transforming data for analytics. It requires minimal development effort compared to the other options, which involve writing and managing custom code or scripts. Glue Crawlers automatically discover the schema, and ETL jobs can be configured visually or with code to convert the CSV files to Parquet and store them in the transformed data bucket. This aligns with the principle of least development effort.

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Question 20

Not Attempted

Storage



A company has 700 TB of backup data stored in network attached storage (NAS) in its data center. This backup data needs to be accessible for infrequent regulatory requests and must be retained for 7 years. The company has decided to migrate this backup data from its data center to AWS. The migration must be complete within 1 month. The company has 500 Mbps of dedicated bandwidth on its public internet connection available for data transfer. What should a solutions architect do to migrate and store the data at the LOWEST cost?

- ✔ Order AWS Snowball devices to transfer the data. Use a lifecycle policy to transition the files to Amazon S3 Glacier Deep Archive.

This option offers the lowest cost and meets the timeline by using Snowball for initial transfer and Glacier Deep Archive for long-term, infrequent access. Given the limited bandwidth, network transfer is not feasible.

- ⊗ Deploy a VPN connection between the data center and Amazon VPC. Use the AWS CLI to copy the data from on premises to Amazon S3 Glacier.

Transferring 700TB over a 500Mbps connection will take significantly longer than one month. Additionally, copying directly to Glacier is not supported and will incur higher costs during the initial transfer. This option is not cost-effective or feasible.

- ⊗ Provision a 500 Mbps AWS Direct Connect connection and transfer the data to Amazon S3. Use a lifecycle policy to transition the files to Amazon S3 Glacier Deep Archive.

Provisioning a Direct Connect connection adds significant cost and won't solve the transfer bandwidth limitation within the 1-month timeframe. Although the lifecycle policy is a good practice, the initial transfer time is a bottleneck.

- ⊗ Use AWS DataSync to transfer the data and deploy a DataSync agent on premises. Use the DataSync task to copy files from the on-premises NAS storage to Amazon S3 Glacier.

DataSync can be a cost-effective way to migrate data, but it won't be fast enough given the 500Mbps bandwidth. Also, you cannot copy directly to S3 Glacier with DataSync. It should be S3 standard first, then configuration S3 Lifecycle to transit to Glacier.

Overall Explanation:

Given the 700TB data size, the 500Mbps bandwidth limitation, and the 1-month deadline, ordering AWS Snowball devices for data transfer is the most cost-effective and feasible solution. After the data is in S3, a lifecycle policy can automatically transition the data to Amazon S3 Glacier Deep Archive for long-term, low-cost storage. Options B, C and D would exceed the 1 month limit for data transfer.

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Question 21

Not Attempted

Storage



A company has a serverless website with millions of objects in an Amazon S3 bucket. The company uses the S3 bucket as the origin for an Amazon CloudFront distribution. The company did not set encryption on the S3 bucket before the objects were loaded. A solutions architect needs to enable encryption for all existing objects and for all objects that are added to the S3 bucket in the future. Which solution will meet these requirements with the LEAST amount of effort?

- ⊗ Create a new S3 bucket. Turn on the default encryption settings for the new S3 bucket. Download all existing objects to temporary local storage. Upload the objects to the new S3 bucket.

This solution requires significant effort to transfer millions of objects, making it a less efficient solution.

- ✓ Turn on the default encryption settings for the S3 bucket. Use the S3 Inventory feature to create a .csv file that lists the unencrypted objects. Run an S3 Batch Operations job that uses the copy command to encrypt those objects.

Enabling default encryption ensures new objects are encrypted. S3 Inventory and Batch Operations provide the least effort to encrypt existing objects.

- ⊗ Create a new encryption key by using AWS Key Management Service (AWS KMS). Change the settings on the S3 bucket to use server-side encryption with AWS KMS managed encryption keys (SSE-KMS). Turn on versioning for the S3 bucket.

Enabling SSE with AWS KMS only applies to new objects. Existing objects would still be unencrypted. Turning on versioning is not required.

- ⊗ Navigate to Amazon S3 in the AWS Management Console. Browse the S3 bucket's objects. Sort by the encryption field. Select each unencrypted object. Use the Modify button to apply default encryption settings to every unencrypted object in the S3 bucket.

Manually modifying each object is impractical for a large number of objects.

Overall Explanation:

The best solution is to turn on default encryption for new objects. To encrypt existing objects, use S3 Inventory to create a list of unencrypted objects, and then use S3 Batch Operations to encrypt them. This minimizes effort compared to the other options.

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Question 22

Not Attempted

Disaster Recovery



A company runs a global web application on Amazon EC2 instances behind an Application Load Balancer. The application stores data in Amazon Aurora. The company needs to create a disaster recovery solution and can tolerate up to 30 minutes of downtime and potential data

loss. The solution does not need to handle the load when the primary infrastructure is healthy. What should a solutions architect do to meet these requirements?

- ☒ Deploy the application with the required infrastructure elements in place. Use Amazon Route 53 to configure active-passive failover. Create an Aurora Replica in a second AWS Region.

This solution provides a standby environment in another region. The active-passive configuration ensures that resources are only used when the primary region fails. The Aurora replica facilitates data replication. Route 53 handles the failover.

- ☐ Host a scaled-down deployment of the application in a second AWS Region. Use Amazon Route 53 to configure active-active failover. Create an Aurora Replica in the second Region.

Active-active failover is not required as the application doesn't need to handle load when the primary is healthy. This option is also more expensive.

- ☐ Replicate the primary infrastructure in a second AWS Region. Use Amazon Route 53 to configure active-active failover. Create an Aurora database that is restored from the latest snapshot.

Restoring from snapshot may take longer than the tolerated 30 minutes of downtime. Active-active failover is not required.

- ☐ Back up data with AWS Backup. Use the backup to create the required infrastructure in a second AWS Region. Use Amazon Route 53 to configure active-passive failover. Create an Aurora second primary instance in the second Region.

Using AWS Backup and restoring might exceed the 30-minute downtime tolerance. Also, an Aurora Replica is sufficient, not a second primary instance.

Overall Explanation:

The best approach is to use an active-passive failover with an Aurora Replica in another region. This meets the requirements of 30 minutes downtime tolerance and minimal cost since the secondary infrastructure is only used in the event of a failure. Option A provides the most appropriate solution by deploying the application infrastructure and utilizing Route 53 for active-passive failover alongside an Aurora Replica.

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Question 23

Not Attempted

Networking



A company hosts a web application on an Amazon EC2 instance within a public subnet. This instance is associated with an Elastic IP address and uses the default security group. However, the default network ACL has been modified to explicitly deny all traffic. A solutions architect is tasked with ensuring the web application is accessible over HTTPS (port 443) from any internet location. Which combination of steps will accomplish this goal?

- ☒ Create a security group rule permitting inbound TCP traffic on port 443 from source 0.0.0.0/0.
This allows external traffic to reach the EC2 instance on the required port. Security groups act as a virtual firewall at the instance level.
- ☐ Create a security group rule permitting outbound TCP traffic on port 443 to destination 0.0.0.0/0.
This option is incorrect because security groups are stateful and only inbound rules are necessary for this scenario. The outbound traffic will be allowed as a response to inbound traffic.
- ☐ Update the network ACL to allow inbound TCP traffic on port 443 from source 0.0.0.0/0.
This is not sufficient on its own. Network ACLs are stateless, requiring explicit rules for both inbound and outbound traffic.
- ☐ Update the network ACL to allow both inbound and outbound TCP traffic on port 443 from/to source/destination 0.0.0.0/0.
This option would allow inbound traffic on port 443, but it would likely block the return traffic from the server since the client's ephemeral ports aren't being accounted for.
- ☒ Update the network ACL to allow inbound TCP traffic on port 443 from source 0.0.0.0/0 and outbound TCP traffic on ephemeral ports 32768-65535 to destination 0.0.0.0/0.
This allows inbound traffic on port 443 and allows the instance to respond to clients by opening up the ephemeral port range for outbound traffic. Network ACLs are stateless, so outbound rules are required.

Overall Explanation:

To enable HTTPS access, we need to configure both the security group and the network ACL. The security group needs an inbound rule allowing TCP traffic on port 443 from 0.0.0.0/0. The network ACL, being stateless, needs both an inbound rule allowing TCP traffic on port 443 from 0.0.0.0/0, and an outbound rule allowing TCP traffic on ephemeral ports

(32768-65535) to 0.0.0.0/0. This allows the EC2 instance to receive HTTPS requests and send responses back to clients on the internet. The ephemeral port range is used for the client's response traffic.

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Question 24

Not Attempted

[Compute](#)

A company's application is experiencing performance degradation. The application is stateful and requires in-memory task completion on Amazon EC2 instances. The infrastructure was initially deployed using AWS CloudFormation with M5 EC2 instances. As traffic has increased, users are reporting delays when accessing the application. What is the MOST operationally efficient solution to address these performance issues?

- ☐

Replace the EC2 instances with T3 EC2 instances that run in an Auto Scaling group. Make the changes by using the AWS Management Console.

T3 instances might not provide sufficient memory capacity for memory-intensive tasks. Using the AWS Management Console is not operationally efficient compared to using CloudFormation templates.
- ☐

Modify the CloudFormation templates to run the EC2 instances in an Auto Scaling group. Increase the desired capacity and the maximum capacity of the Auto Scaling group manually when an increase is necessary.

While using Auto Scaling can help with scaling, manually adjusting capacity does not provide the most efficient automated solution, and does not address the memory constraints of the application.
- ☐

Modify the CloudFormation templates. Replace the EC2 instances with R5 EC2 instances. Use Amazon CloudWatch built-in EC2 memory metrics to track the application performance for future capacity planning.

R5 instances are memory-optimized. However, built-in EC2 memory metrics in CloudWatch don't provide granular insights into application performance and require a custom agent.
- ☒

Modify the CloudFormation templates. Replace the EC2 instances with R5 EC2 instances. Deploy the Amazon CloudWatch agent on the EC2 instances to generate custom application latency metrics for future capacity planning.

R5 instances provide better memory optimization, and deploying the CloudWatch agent allows for generating custom application latency metrics, offering valuable insight into application performance and enables future capacity planning. This is the most operationally efficient method

because it combines a memory optimized instance type with custom monitoring for better capacity planning.

Overall Explanation:

The most operationally efficient solution is to migrate to R5 instances, which are optimized for memory-intensive workloads. Additionally, deploying the CloudWatch agent allows for the generation of custom application latency metrics, providing more in-depth monitoring and capacity planning capabilities. EC2 instances don't provide default memory metrics to CloudWatch without the agent.

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Question 25

Not Attempted

Storage



A company is developing a media-sharing application using Amazon S3 for object storage. They require high availability and resilience to Availability Zone failures. The application will store both frequently accessed hot media and infrequently accessed cold media, with access patterns that are unpredictable. To optimize costs, the storage solution must automatically transition objects between storage tiers based on access frequency. Which S3 storage class best meets these requirements?

☐ S3 Standard

S3 Standard is suitable for frequently accessed data, but not cost-effective for rarely accessed data as it doesn't automatically tier based on access patterns.

☒ S3 Intelligent-Tiering

S3 Intelligent-Tiering automatically moves data to the most cost-effective tier based on access patterns, making it ideal for unpredictable access patterns. It also provides high availability by storing data across multiple Availability Zones.

☐ S3 Standard-Infrequent Access (S3 Standard-IA)

S3 Standard-IA is suitable for infrequently accessed data but requires manual or lifecycle policy-based transitions, making it less ideal for unpredictable access patterns.

☐ S3 One Zone-Infrequent Access (S3 One Zone-IA)

S3 One Zone-IA stores data in a single Availability Zone, so it is not resilient to Availability Zone failures, and it doesn't handle frequently accessed data cost-effectively.

Overall Explanation:

S3 Intelligent-Tiering is designed for scenarios where data access patterns are unpredictable. It automatically moves data between frequent, infrequent, and archive access tiers to optimize costs while maintaining performance. It also stores data redundantly across multiple Availability Zones providing high availability. S3 Standard is not cost-effective for infrequently accessed data. S3 Standard-IA requires defining lifecycle rules and isn't as dynamic. S3 One Zone-IA doesn't provide cross-AZ redundancy.

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Question 26

Not Attempted

Serverless



A solutions architect is designing a new API using Amazon API Gateway that will receive requests from users. The volume of requests is highly variable; several hours can pass without receiving a single request. The data processing will take place asynchronously, but should be completed within a few seconds after a request is made. Which compute service should the solutions architect have the API invoke to deliver the requirements at the lowest cost?

☐ An AWS Glue job

AWS Glue is designed for ETL tasks and is not suitable for handling API requests with low latency requirements. It is also not the most cost-effective option for variable request volumes.

☒ An AWS Lambda function

AWS Lambda is a serverless compute service ideal for processing asynchronous requests triggered by API Gateway. It scales automatically and charges only for compute time, making it the most cost-effective solution for variable request volumes and low latency requirements.

☐ A containerized service hosted in Amazon Elastic Kubernetes Service (Amazon EKS)

While EKS offers scalability, it introduces overhead and complexity in managing and scaling infrastructure, making it less cost-effective for intermittent API requests.

☐ A containerized service hosted in Amazon ECS with Amazon EC2

Similar to EKS, using ECS with EC2 involves infrastructure management and scaling, which is not as cost-effective or efficient as Lambda for variable API request volumes.

Overall Explanation:

AWS Lambda is the best choice because it is a serverless compute service that can be triggered by API Gateway to process requests asynchronously. It automatically scales based on the incoming request volume and allows for cost optimization by charging only for the actual compute time used to process the requests. The other options involve more infrastructure management and are less cost-effective for variable request volumes. The data processing needs to be completed within a few seconds, which Lambda can easily achieve.

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Question 27

Not Attempted

Storage



A company runs an application on a group of Amazon Linux EC2 instances. For compliance reasons, the company must retain all application log files for 7 years. The log files will be analyzed by a reporting tool that must be able to access all the files concurrently. Which storage solution meets these requirements MOST cost-effectively?

- ☐ Amazon Elastic Block Store (Amazon EBS)
EBS provides block-level storage for EC2 instances. It is not the most cost-effective for long-term log retention, and does not inherently provide concurrent access required for the reporting tool.
- ☐ Amazon Elastic File System (Amazon EFS)
EFS provides scalable file storage accessible by multiple EC2 instances concurrently. While meeting the concurrent access requirement, it is generally more expensive than S3 for long-term storage of log files.
- ☐ Amazon EC2 instance store
Instance store is temporary storage directly attached to an EC2 instance. It is not durable, does not meet the 7-year retention requirement, and is not accessible outside of the specific EC2 instance.
- ☒ Amazon S3
Amazon S3 offers cost-effective object storage suitable for long-term retention of log files. It supports concurrent access for analysis by the reporting tool. S3 is generally the most cost-effective option for the described scenario.

Overall Explanation:

Amazon S3 is the most cost-effective solution for long-term storage and concurrent access to log files. EBS is not cost-effective, EFS is more expensive than S3, and instance store is not durable or accessible for the given requirements.

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Question 28

Not Attempted

Security



A company has hired an external vendor to perform work in the company's AWS account. The vendor uses an automated tool that is hosted in an AWS account that the vendor owns. The vendor does not have IAM access to the company's AWS account. How should a solutions architect grant this access to the vendor?

- ☒ Create an IAM role in the company's account to delegate access to the vendor's IAM role. Attach the appropriate IAM policies to the role for the permissions that the vendor requires.

This is the correct approach. By creating an IAM role and delegating access to the vendor's IAM role, you establish a trust relationship between accounts. This allows the vendor's automated tool to assume the role in the company's account and access the necessary resources. Attaching the appropriate IAM policies to the role defines the precise permissions that the vendor requires.

- ☐ Create an IAM user in the company's account with a password that meets the password complexity requirements. Attach the appropriate IAM policies to the user for the permissions that the vendor requires.

This is incorrect because creating an IAM user with a password would require sharing the credentials with the vendor, which is not recommended for security reasons.

- ☐ Create an IAM group in the company's account. Add the tool's IAM user from the vendor account to the group. Attach the appropriate IAM policies to the group for the permissions that the vendor requires.

This is incorrect because adding the vendor's IAM user to an IAM group in the company's account would not provide a direct and controlled way to delegate access to the vendor's tool.

- ☐ Create a new identity provider by choosing "AWS account" as the provider type in the IAM console. Supply the vendor's AWS account ID and user name. Attach the appropriate IAM policies to the new provider for the permissions that the vendor requires.

This is incorrect because creating a new identity provider for the vendor's AWS account would not provide a straightforward way to delegate access to the vendor's tool. Identity providers are typically used for federated access using external identity systems.

Overall Explanation:

The best solution is to create an IAM role in the company's account to delegate access to the vendor's IAM role. This establishes a trust relationship and allows the vendor's tool to assume the role and access resources with defined permissions. Creating IAM users or using Identity Providers are less secure or less appropriate methods for this scenario.

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Question 29

Not Attempted

Networking



A company has deployed a Java Spring Boot application as a pod running on Amazon Elastic Kubernetes Service (Amazon EKS) in private subnets. The application needs to write data to an Amazon DynamoDB table. A solutions architect must ensure the application can interact with the DynamoDB table without exposing traffic to the internet. Which combination of steps should the solutions architect take to accomplish this goal? (Choose two.)

- ☒ **Attach an IAM role that has sufficient privileges to the EKS pod.**
Attaching an IAM role to the EKS pod grants the necessary permissions for the pod to access DynamoDB. The IAM role should have appropriate policies allowing write access to the DynamoDB table.
- ☐ **Attach an IAM user that has sufficient privileges to the EKS pod.**
Attaching an IAM user to the pod is not a recommended approach for EKS. IAM users are meant for individual access and not service-to-service authentication.
- ☐ **Allow outbound connectivity to the DynamoDB table through the private subnets' network ACLs.**
Configuring outbound connectivity through network ACLs alone will not provide a secure and direct connection to DynamoDB without exposing traffic to the internet.

☒ Create a VPC endpoint for DynamoDB.

Creating a VPC endpoint for DynamoDB allows the EKS pod to access DynamoDB privately within the VPC, without the need for internet connectivity. The VPC endpoint provides a direct and secure connection to DynamoDB.

☐ Embed the access keys in the Java Spring Boot code.

Embedding access keys in the code is not a recommended security practice. It can lead to potential security vulnerabilities.

Overall Explanation:

To securely access DynamoDB from an EKS pod in private subnets without exposing traffic to the internet, you need to attach an IAM role to the EKS pod that grants write permissions to the DynamoDB table. Additionally, creating a VPC endpoint for DynamoDB allows the EKS pod to access DynamoDB privately within the VPC. Therefore, options A and D are the correct answers.

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Question 30

Not Attempted

Networking



A company recently migrated its web application to AWS by rehosting the application on Amazon EC2 instances in a single AWS Region. The company wants to redesign its application architecture to be highly available and fault tolerant. Traffic must reach all running EC2 instances randomly. Which combination of steps should the company take to meet these requirements?

☐ Create an Amazon Route 53 failover routing policy.

Failover routing is designed to direct traffic to a backup resource or secondary location only when the primary resource or location is unavailable. It doesn't provide random distribution of traffic.

☐ Create an Amazon Route 53 weighted routing policy.

Although a weighted routing policy allows you to distribute traffic across multiple EC2 instances, it does not ensure an equal and random distribution without proper configuration. Weighted routing distributes traffic based on weights assigned to each instance, not necessarily randomly.

☒ Create an Amazon Route 53 multivalue answer routing policy.

A multivalue answer routing policy in Route 53 allows you to configure multiple values for a DNS record, and Route 53 responds to DNS queries with multiple random values. This enables the distribution of traffic randomly among the available EC2 instances. It can return up to 8 healthy records at random.

- ⊗ Launch three EC2 instances: two instances in one Availability Zone and one instance in another Availability Zone.

While launching instances in multiple AZs is important for fault tolerance, having only three instances does not provide an optimal distribution for fault tolerance. With only three instances, the traffic distribution and redundancy may not be evenly distributed, potentially leading to imbalanced resource utilization.

- ✓ Launch four EC2 instances: two instances in one Availability Zone and two instances in another Availability Zone.

By launching EC2 instances in different AZs, you achieve high availability and fault tolerance. Launching four instances (two in each AZ) ensures that there are enough resources to handle the traffic load and maintain the desired level of availability across multiple availability zones.

Overall Explanation:

To achieve high availability, fault tolerance, and random traffic distribution across EC2 instances, the company should use a Route 53 multivalue answer routing policy (option C) in conjunction with launching four EC2 instances across two Availability Zones (option E). The multivalue answer policy provides a random selection of healthy instances, while distributing instances across AZs ensures that the application can withstand the failure of one AZ. Weighted routing could work, but requires careful configuration of weights, whereas Multi Value, if all health checks are healthy, will distribute records evenly.

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Question 31

Not Attempted

Databases



A media company collects and analyzes user activity data on premises. The company wants to migrate this capability to AWS. The user activity data store will continue to grow and will be petabytes in size. The company needs to build a highly available data ingestion solution that facilitates on-demand analytics of existing data and new data with SQL. Which solution will meet these requirements with the LEAST operational overhead?

- ⊗ Send activity data to an Amazon Kinesis data stream. Configure the stream to deliver the data to an Amazon S3 bucket.

Using Kinesis Data Streams requires more manual intervention for scaling and data processing. It also requires additional services to perform analytics on data stored in S3, increasing operational overhead.

- ✔ Send activity data to an Amazon Kinesis Data Firehose delivery stream. Configure the stream to deliver the data to an Amazon Redshift cluster.

Kinesis Data Firehose simplifies the delivery of streaming data to Amazon Redshift, a petabyte-scale data warehouse, enabling on-demand SQL analytics with minimal operational overhead. Both services are fully managed, handling scalability, availability, and durability.

- ⊗ Place activity data in an Amazon S3 bucket. Configure Amazon S3 to run an AWS Lambda function on the data as the data arrives in the S3 bucket.

Using Lambda for data transformation as data arrives in S3 adds operational overhead and doesn't directly facilitate on-demand SQL analytics without additional services.

- ⊗ Create an ingestion service on Amazon EC2 instances that are spread across multiple Availability Zones. Configure the service to forward data to an Amazon RDS Multi-AZ database.

Creating a custom ingestion service on EC2 instances and forwarding data to an RDS database requires significant operational overhead for managing the EC2 instances, the ingestion service, and the database. RDS is not suitable for petabyte-scale data.

Overall Explanation:

The most efficient solution is to use Amazon Kinesis Data Firehose to ingest data directly into Amazon Redshift. Firehose is a fully managed service that simplifies data ingestion and scaling. Redshift is a petabyte-scale data warehouse designed for on-demand SQL analytics. This combination minimizes operational overhead.

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Question 32

Not Attempted

Serverless



A company collects data from thousands of remote devices using a RESTful web services application hosted on an Amazon EC2 instance. The EC2 instance receives, transforms, and stores all data in an Amazon S3 bucket. The number of remote devices will soon increase to

millions. The company requires a highly scalable solution that minimizes operational overhead. Which combination of actions should a solutions architect take to meet these requirements? (Choose two.)

- ☒ Use AWS Glue to process the raw data in Amazon S3.
AWS Glue can automatically discover the schema of the data in S3 and generate ETL code to transform it, providing a scalable and managed data processing solution.
- ☐ Use Amazon Route 53 to route traffic to different EC2 instances.
Amazon Route 53 primarily focuses on DNS management and routing traffic to different endpoints, but it does not directly address the need for scalable data processing.
- ☐ Add more EC2 instances to accommodate the increasing amount of incoming data.
Adding more EC2 instances can lead to increased operational overhead in terms of managing and scaling the instances, and it's not the most cost-effective or scalable solution.
- ☐ Send the raw data to Amazon Simple Queue Service (Amazon SQS). Use EC2 instances to process the data.
Using SQS and EC2 for processing data introduces more complexity and operational overhead compared to serverless alternatives.
- ☒ Use Amazon API Gateway to send the raw data to an Amazon Kinesis data stream. Configure Amazon Kinesis Data Firehose to use the data stream as a source to deliver the data to Amazon S3.
API Gateway provides a scalable and managed infrastructure to handle incoming requests from remote devices. Kinesis Data Streams can ingest the data, and Kinesis Data Firehose can deliver the data to S3.

Overall Explanation:

The best solution involves using API Gateway to handle the incoming data stream and Kinesis Data Firehose to deliver that data stream to S3. Also, AWS Glue can then process the raw data once it arrives in S3. This provides a scalable serverless solution. Route 53 does not directly address the data processing needs. Adding more EC2 instances will increase operational costs. SQS and EC2 introduce complexity and operational overhead.

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Question 33

Not Attempted

Storage



A company needs to retain its AWS CloudTrail logs for 3 years. The company is enforcing CloudTrail across a set of AWS accounts by using AWS Organizations from the parent account. The CloudTrail target S3 bucket is configured with S3 Versioning enabled. An S3 Lifecycle policy is in place to delete *current* objects after 3 years. After the fourth year of use of the S3 bucket, the S3 bucket metrics show that the number of objects has continued to rise. However, the number of new CloudTrail logs that are delivered to the S3 bucket has remained consistent. Which solution will delete objects that are older than 3 years in the *MOST* cost-effective manner?

- ☐ ⓧ Configure the organization's centralized CloudTrail trail to expire objects after 3 years.
This option is not directly related to managing objects in the S3 bucket and focuses on configuring the expiration of CloudTrail trails, which does not address the need to delete existing objects from the S3 bucket.
- ☒ ⓧ Configure the S3 Lifecycle policy to delete previous versions as well as current versions.
Versioning has caused the number of objects to increase over time, even as current objects are deleted after 3 years. By deleting previous versions as well, this will clean up old object versions and reduce storage costs. An S3 Lifecycle policy incurs no additional charges and requires no additional resources to configure and run. It is a native S3 tool for managing object lifecycles cost-effectively.
- ☐ ⓧ Create an AWS Lambda function to enumerate and delete objects from Amazon S3 that are older than 3 years.
While technically possible, this approach introduces additional complexity and operational overhead compared to using S3 Lifecycle policies.
- ☐ ⓧ Configure the parent account as the owner of all objects that are delivered to the S3 bucket.
Changing the ownership of the objects in the S3 bucket does not directly address the need to delete objects older than 3 years. Ownership does not affect the deletion behavior of the objects.

Overall Explanation:

The most cost-effective solution is to configure the S3 Lifecycle policy to delete previous versions as well as current versions. The existing lifecycle policy only deletes current versions of objects, but with S3 Versioning enabled, older versions remain in the bucket. By including previous versions in the lifecycle policy, all object versions older than 3 years will

be deleted, reducing storage costs. Using Lambda would add unneeded complexity and cost.

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Question 34

Not Attempted

Databases



A company's API receives real-time data from monitoring devices and stores it in an Amazon RDS DB instance. During peak traffic, the API experiences timeout errors due to the database's inability to handle the write volume. A solutions architect needs to minimize database connections and prevent data loss during high-traffic periods. Which solution meets these requirements?

- ☐ Increase the size of the DB instance to an instance type with more available memory.
Increasing the DB instance size might improve performance but doesn't directly address connection limits or prevent data loss during traffic spikes.
- ☐ Modify the DB instance to be a Multi-AZ DB instance. Configure the application to write to all active RDS DB instances.
Multi-AZ improves availability but doesn't solve the problem of managing high write traffic and minimizing database connections. Writing to multiple instances doesn't help manage the initial surge.
- ☒ Modify the API to write incoming data to an Amazon Simple Queue Service (Amazon SQS) queue. Use an AWS Lambda function that Amazon SQS invokes to write data from the queue to the database.
This solution decouples the API from the database, using SQS as a buffer to handle write traffic. The Lambda function processes data from the queue and writes it to the database, minimizing connections and preventing data loss during peaks.
- ☐ Modify the API to write incoming data to an Amazon Simple Notification Service (Amazon SNS) topic. Use an AWS Lambda function that Amazon SNS invokes to write data from the topic to the database.
SNS is designed for fan-out scenarios and not ideal for queuing data to be reliably written to a database. SQS is better suited for this purpose.

Overall Explanation:

The optimal solution is to use an Amazon SQS queue to buffer incoming data from the API. An AWS Lambda function, triggered by the SQS queue, then writes the data to the database. This approach minimizes database connections, prevents data loss during heavy traffic, and decouples the API from the database, making the system more resilient. While SNS with Lambda can provide decoupling, it's less suited for guaranteeing data delivery in this scenario compared to SQS.

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Question 35

Not Attempted

Databases



A company manages its own Amazon EC2 instances that run MySQL databases. The company is manually managing replication and scaling as demand increases or decreases. The company needs a new solution that simplifies the process of adding or removing compute capacity to or from its database tier as needed. The solution also must offer improved performance, scaling, and durability with minimal effort from operations.

☒ **Migrate the databases to Amazon Aurora Serverless for Aurora MySQL.**

Migrating the databases to Aurora Serverless provides automated scaling and replication capabilities. Aurora Serverless automatically scales the capacity based on the workload, allowing for seamless addition or removal of compute capacity as needed. It also offers improved performance, durability, and high availability without requiring manual management of replication and scaling.

☐ **Migrate the databases to Amazon Aurora Serverless for Aurora PostgreSQL.**

Migrating to a different database engine (PostgreSQL) may introduce compatibility issues and require significant code modifications.

☐ **Combine the databases into one larger MySQL database. Run the larger database on larger EC2 instances.**

Consolidating into a larger MySQL database on larger EC2 instances does not provide the desired scalability and automation.

☐ **Create an EC2 Auto Scaling group for the database tier. Migrate the existing databases to the new environment.**

Using EC2 Auto Scaling groups for the database tier still requires manual management of replication and scaling.

Overall Explanation:

S3 Glacier Deep Archive is the most cost-effective solution for storing backup files that are frequently accessed for 1 month and rarely accessed after that, while still needing to be kept indefinitely. An S3 Lifecycle configuration automates the process of transitioning the objects to Glacier Deep Archive after one month, minimizing storage costs. Therefore Option B is the correct option.

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Question 37

Not Attempted

Networking



A company is concerned that two NAT instances in use will no longer be able to support the traffic needed for the company's application. A solutions architect wants to implement a solution that is highly available, fault tolerant, and automatically scalable. Which of the following should the solutions architect recommend?

- ☐ Remove the two NAT instances and replace them with two NAT gateways in the same Availability Zone.
Placing both NAT gateways in the same Availability Zone does not provide fault tolerance.
- ☐ Use Auto Scaling groups with Network Load Balancers for the NAT instances in different Availability Zones.
Using Auto Scaling groups with Network Load Balancers is not the recommended approach for NAT instances. NAT Gateways are the managed service that should be used for HA, Fault Tolerance and Scalability
- ☒ Remove the two NAT instances and replace them with two NAT gateways in different Availability Zones.
This recommendation ensures high availability and fault tolerance by distributing the NAT gateways across multiple AZs. NAT gateways are managed AWS services that provide scalable and highly available outbound NAT functionality. By deploying NAT gateways in different AZs, the company can achieve redundancy and avoid a single point of failure. This solution also provides automatic scaling to handle increasing traffic without manual intervention.
- ☐ Replace the two NAT instances with Spot Instances in different Availability Zones and deploy a Network Load Balancer.
Spot Instances are not suitable for critical infrastructure components like NAT instances.

Overall Explanation:

The best solution is to replace the NAT instances with two NAT gateways in different Availability Zones. This provides high availability, fault tolerance, and automatic scalability, as NAT gateways are a managed service that scales automatically and are designed to be highly available. Placing them in different AZs provides redundancy. Using NAT instances with autoscaling groups and NLBs is more complex and less reliable than using NAT Gateways. Spot instances are also not suitable for this scenario because they can be terminated at any time, which could disrupt the application's internet connectivity.

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Question 38

Not Attempted

Networking



An application runs on an Amazon EC2 instance that has an Elastic IP address in VPC A. The application requires access to a database in VPC B. Both VPCs are in the same AWS account. Which solution will provide the required access MOST securely?

- ☐ Create a DB instance security group that allows all traffic from the public IP address of the application server in VPC A.

This option involves exposing the database to the internet (via a public IP) and trusting traffic based on the source IP. This is less secure than using private networking.

- ☒ Configure a VPC peering connection between VPC A and VPC B.

VPC peering establishes a private network connection between the two VPCs, allowing secure communication without exposing resources to the public internet. This is the most secure option.

- ☐ Make the DB instance publicly accessible. Assign a public IP address to the DB instance. Making the DB instance publicly accessible is highly insecure and should be avoided.

- ☐ Launch an EC2 instance with an Elastic IP address into VPC B. Proxy all requests through the new EC2 instance.

This option adds unnecessary complexity and cost. It doesn't address the security concern as effectively as VPC peering.

Overall Explanation:

EC2 Instance State-change Notifications are triggered by instance state transitions (e.g., starting, stopping). RDP or SSH access does not cause an instance state change, so this solution would not provide the required notifications.

Overall Explanation:

Publishing VPC flow logs to CloudWatch Logs, creating metric filters to detect RDP or SSH access, and configuring a CloudWatch metric alarm is the most effective way to notify the operations team when RDP or SSH access is established. Options A, B, and D do not directly address the requirement of detecting and notifying based on RDP/SSH access patterns.

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Question 40

Not Attempted

Security



A solutions architect has created a new AWS account and must secure AWS account root user access. Which combination of actions will accomplish this? (Choose two.)

- ☒ Ensure the root user uses a strong password.
Setting a strong password for the root user is an essential security measure to prevent unauthorized access.
- ☒ Enable multi-factor authentication to the root user.
Enabling MFA adds an extra layer of security by requiring an additional authentication factor, such as a code from a mobile app or a hardware token, in addition to the password.
- ☐ Store root user access keys in an encrypted Amazon S3 bucket.
Root user access keys should be avoided whenever possible, and it is best to use IAM users with restricted permissions instead.
- ☐ Add the root user to a group containing administrative permissions.
The root user already has unrestricted access to all resources and services in the account, so granting additional administrative permissions could increase the risk of unauthorized actions.
- ☐ Apply the required permissions to the root user with an inline policy document.
It is recommended to create IAM users with appropriate permissions and use those users for day-to-day operations, while keeping the root user secured and only using it for necessary administrative tasks.

Overall Explanation:

Securing the root user involves using a strong password and enabling multi-factor authentication (MFA). Avoid storing root user access keys or granting additional permissions to the root user, as it already has full administrative access. Best practice suggests creating IAM users for day-to-day operations and reserving the root user for essential tasks only.

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Question 41

Not Attempted

Security



A company is building a new web-based customer relationship management application. The application will use several Amazon EC2 instances that are backed by Amazon Elastic Block Store (Amazon EBS) volumes behind an Application Load Balancer (ALB). The application will also use an Amazon Aurora database. All data for the application must be encrypted at rest and in transit. Which solution will meet these requirements?

- ☐ Use AWS Key Management Service (AWS KMS) certificates on the ALB to encrypt data in transit. Use AWS Certificate Manager (ACM) to encrypt the EBS volumes and Aurora database storage at rest.

This option is incorrect because it reverses the roles of ACM and KMS. ACM is not used to encrypt EBS volumes and Aurora database storage at rest.

- ☐ Use the AWS root account to log in to the AWS Management Console. Upload the company's encryption certificates. While in the root account, select the option to turn on encryption for all data at rest and in transit for the account.

This option is incorrect because relying on the AWS root account and a single "turn on encryption" option is not a valid or secure approach.

- ☒ Use AWS Key Management Service (AWS KMS) to encrypt the EBS volumes and Aurora database storage at rest. Attach an AWS Certificate Manager (ACM) certificate to the ALB to encrypt data in transit.

This option is correct. KMS encrypts EBS volumes and Aurora storage at rest, and ACM certificates on the ALB encrypt data in transit.

- ☐ Use BitLocker to encrypt all data at rest. Import the company's TLS certificate keys to AWS Key Management Service (AWS KMS) Attach the KMS keys to the ALB to encrypt

data in transit.

This option is incorrect because BitLocker is not a suitable solution for encrypting data on AWS services. Also, importing TLS certificate keys to KMS and attaching them to ALB is not standard practice.

Overall Explanation:

The correct solution involves using AWS KMS to encrypt data at rest for EBS volumes and the Aurora database. For encrypting data in transit, an AWS Certificate Manager (ACM) certificate should be attached to the Application Load Balancer (ALB) to enable HTTPS and TLS encryption.

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Question 42

Not Attempted

Databases



A company is migrating its on-premises Oracle database to Amazon Aurora PostgreSQL. Several applications write to the same database tables. These applications will be migrated one by one, with a month separating each migration. Management is concerned about the database's high read/write activity. The data must remain synchronized across both databases throughout the migration process. Which solution should a solutions architect recommend?

- ☐ Use AWS DataSync for the initial migration. Use AWS Database Migration Service (AWS DMS) to create a change data capture (CDC) replication task and a table mapping to select all tables.

AWS DataSync is suitable for file transfer, not database migration. It does not handle schema conversion or continuous synchronization needed for this scenario.

- ☐ Use AWS DataSync for the initial migration. Use AWS Database Migration Service (AWS DMS) to create a full load plus change data capture (CDC) replication task and a table mapping to select all tables.

AWS DataSync is suitable for file transfer, not database migration. It does not handle schema conversion required for migrating from Oracle to PostgreSQL.

- ☒ Use the AWS Schema Conversion Tool with AWS Database Migration Service (AWS DMS) using a memory optimized replication instance. Create a full load plus change data capture (CDC) replication task and a table mapping to select all tables.

This option correctly utilizes AWS SCT for schema conversion and AWS DMS for data migration and synchronization. Using a memory-optimized instance addresses the high read/write requirements, and selecting all tables ensures complete data consistency.

- ⊗ Use the AWS Schema Conversion Tool with AWS Database Migration Service (AWS DMS) using a compute optimized replication instance. Create a full load plus change data capture (CDC) replication task and a table mapping to select the largest tables.

While AWS SCT and DMS are appropriate, a compute-optimized instance is less suitable for databases with high read/write activity. Mapping only the largest tables will lead to data inconsistencies and application failures.

Overall Explanation:

The correct answer is C. This approach uses AWS Schema Conversion Tool (SCT) to convert the Oracle schema to Aurora PostgreSQL. It utilizes AWS Database Migration Service (DMS) with a memory-optimized replication instance due to the high read/write activity. The Full load plus CDC replication task ensures continuous data synchronization and mapping all tables guarantees complete migration. DataSync is not designed for database migrations, and compute-optimized instance is less suitable for high-read/write database activity and only mapping the largest tables results to data inconsistencies.

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Question 43

Not Attempted

Databases



A company hosts an image sharing application utilizing a three-tier architecture. The front-end, application, and database layers are each deployed on individual Amazon EC2 instances. The database layer uses MySQL. A solutions architect needs to design a solution that provides scalability and high availability with minimal code changes. Which of the following options best satisfies these requirements?

- ⊗ Use Amazon S3 to host the front-end layer. Use AWS Lambda functions for the application layer. Migrate the database to Amazon DynamoDB. Use Amazon S3 to store and serve users' images.

This solution involves significant architectural changes, requiring substantial code modifications, especially for the database migration to DynamoDB and the application layer to Lambda. S3 is a good choice for image storage, but the other changes are too extensive.

- ⊗ Deploy load-balanced Multi-AZ AWS Elastic Beanstalk environments for the front-end and application layers. Migrate the database to an Amazon RDS DB instance with multiple read replicas to serve users' images.

Elastic Beanstalk provides scalability and high availability with minimal application changes. However, using read replicas for serving images isn't the most efficient approach for serving images; S3 is a better fit. Read Replicas also do not provide HA for writes.

- ⊗ Use Amazon S3 to host the front-end layer. Use a fleet of EC2 instances in an Auto Scaling group for the application layer. Move the database to a memory-optimized instance type to store and serve users' images.

This solution requires changes to the front-end and application tiers, reducing the amount of changes is preferable. Storing images in a memory optimized database instance is not cost-effective or scalable.

- ✓ Deploy load-balanced Multi-AZ AWS Elastic Beanstalk environments for the front-end and application layers. Migrate the database to an Amazon RDS Multi-AZ DB instance. Use Amazon S3 to store and serve users' images.

Elastic Beanstalk simplifies deployment and management, providing scalability and high availability for the application tiers. RDS Multi-AZ provides database high availability. S3 is the optimal solution for storing and serving images. This option minimizes application changes while meeting the requirements.

Overall Explanation:

The best solution is to use load-balanced Multi-AZ AWS Elastic Beanstalk environments for the front-end and application layers. This minimizes code changes and provides scalability and high availability. Moving the database to an Amazon RDS Multi-AZ DB instance ensures database high availability with automatic failover. Storing images in Amazon S3 provides a highly scalable and durable solution optimized for image storage and delivery.

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Question 44

Not Attempted

Networking



An application running on an Amazon EC2 instance in VPC-A needs to access files in another EC2 instance in VPC-B. Both VPCs are in separate AWS accounts. The network administrator needs to design a solution to configure secure access to the EC2 instance in VPC-B from VPC-A.

The connectivity should not have a single point of failure or bandwidth concerns. Which solution will meet these requirements?

- ☒ Set up a VPC peering connection between VPC-A and VPC-B.
A VPC peering connection allows secure communication between instances in different VPCs using private IP addresses. It doesn't rely on a single point of failure and AWS uses its existing infrastructure for the peering connection, so there isn't a bandwidth bottleneck.
- ☐ Set up VPC gateway endpoints for the EC2 instance running in VPC-B.
VPC gateway endpoints are used for accessing AWS services like S3 or DynamoDB, not for connecting EC2 instances in different VPCs.
- ☐ Attach a virtual private gateway to VPC-B and set up routing from VPC-A.
This would require configuring a VPN connection, which introduces complexity and potential single points of failure, plus extra bandwidth costs and limitations.
- ☐ Create a private virtual interface (VIF) for the EC2 instance running in VPC-B and add appropriate routes from VPC-A.
Private VIFs are used with Direct Connect to connect on-premises infrastructure to VPCs, not for VPC-to-VPC communication within AWS.

Overall Explanation:

VPC peering is the best solution because it establishes a direct, secure network route between VPC-A and VPC-B using private IP addresses. It avoids single points of failure and bandwidth limitations inherent in other approaches like VPNs or gateway endpoints. AWS uses the existing infrastructure of a VPC to create a VPC peering connection; it is neither a gateway nor a VPN connection, and does not rely on a separate piece of physical hardware.

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Question 45

Not Attempted

[Monitoring and Logging](#)

A company wants to experiment with individual AWS accounts for its engineer team. The company requires notifications when the Amazon EC2 instance usage for a given month exceeds a specific threshold for each account. What should a solutions architect do to meet this requirement MOST cost-effectively?

- ⊗ Use Cost Explorer to create a daily report of costs by service. Filter the report by EC2 instances. Configure Cost Explorer to send an Amazon Simple Email Service (Amazon SES) notification when a threshold is exceeded.

Cost Explorer is useful for analyzing costs, but it does not provide the real-time alerting capability that AWS Budgets offers. Daily reports might be too frequent and increase cost.

- ⊗ Use Cost Explorer to create a monthly report of costs by service. Filter the report by EC2 instances. Configure Cost Explorer to send an Amazon Simple Email Service (Amazon SES) notification when a threshold is exceeded.

Cost Explorer is useful for analyzing costs, but it does not provide the real-time alerting capability that AWS Budgets offers.

- ✔ Use AWS Budgets to create a cost budget for each account. Set the period to monthly. Set the scope to EC2 instances. Set an alert threshold for the budget. Configure an Amazon Simple Notification Service (Amazon SNS) topic to receive a notification when a threshold is exceeded.

AWS Budgets allows you to create budgets for your AWS accounts and set alerts when usage exceeds a certain threshold. By creating a budget for each account, specifying the period as monthly and the scope as EC2 instances, you can effectively track the EC2 usage for each account and be notified when a threshold is exceeded. This solution is the most cost-effective option.

- ⊗ Use AWS Cost and Usage Reports to create a report with hourly granularity. Integrate the report data with Amazon Athena. Use Amazon EventBridge to schedule an Athena query. Configure an Amazon Simple Notification Service (Amazon SNS) topic to receive a notification when a threshold is exceeded.

Using AWS Cost and Usage Reports integrated with Amazon Athena and Amazon EventBridge is a more complex and potentially costlier solution compared to AWS Budgets for this specific use case. It's also more suitable for fine-grained, custom analytics rather than straightforward threshold-based alerts.

Overall Explanation:

AWS Budgets is the most cost-effective solution. It allows setting a budget for costs and usage for accounts and setting alerts when the budget threshold is exceeded in real-time. Other options involve more complex and costly services such as Cost Explorer with SES or Cost and Usage Reports with Athena and EventBridge, which are not designed for simple budget alerting.

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Question 46

Not Attempted

Serverless



A solutions architect is designing a new microservice for a company's application. Clients must be able to call an HTTPS endpoint to reach the microservice. The microservice also must use AWS Identity and Access Management (IAM) to authenticate calls. The solutions architect will write the logic for this microservice by using a single AWS Lambda function that is written in Go 1.x. Which solution will deploy the function in the MOST operationally efficient way?

- ☐ Create an Amazon API Gateway REST API. Configure the method to use the Lambda function. Enable IAM authentication on the API.
- While API Gateway can handle HTTPS endpoints and IAM authentication, it introduces more operational overhead than Lambda Function URLs. API Gateway offers additional features like request validation and throttling, which are not mentioned as requirements.
- ☒ Create a Lambda function URL for the function. Specify AWS_IAM as the authentication type.

Lambda function URLs with AWS_IAM authentication provide the most operationally efficient solution. They allow you to create HTTPS endpoints quickly and directly integrate with IAM for authentication, minimizing configuration and operational overhead.

- ☐ Create an Amazon CloudFront distribution. Deploy the function to Lambda@Edge. Integrate IAM authentication logic into the Lambda@Edge function.

Lambda@Edge is more complex and not the most efficient solution for simple HTTPS endpoint and IAM authentication. Additionally, Lambda@Edge does not support Go.

- ☐ Create an Amazon CloudFront distribution. Deploy the function to CloudFront Functions. Specify AWS_IAM as the authentication type.

CloudFront Functions are used for lightweight tasks and do not support IAM authentication directly. This option also adds unnecessary complexity.

Overall Explanation:

The most operationally efficient solution is to create a Lambda function URL and specify AWS_IAM as the authentication type. This option provides a straightforward HTTPS endpoint with built-in IAM authentication, minimizing configuration and operational overhead compared to API Gateway or CloudFront. Lambda@Edge also does not support the Go 1.x runtime specified in the question. Therefore option 2 is the correct answer.

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Question 47

Not Attempted

Monitoring and Logging



A company observes an increase in Amazon EC2 costs in its most recent bill. The billing team notices unwanted vertical scaling of instance types for a couple of EC2 instances. A solutions architect needs to create a graph comparing the last 2 months of EC2 costs and perform an in-depth analysis to identify the root cause of the vertical scaling. Which approach should the solutions architect take to generate the required information with the LEAST operational overhead?

- ☐ Use AWS Budgets to create a budget report and compare EC2 costs based on instance types.

AWS Budgets are primarily for setting spending goals and receiving alerts, not for in-depth cost analysis of past performance based on instance types. Also, it's from the time we create it. We can not see report from the last 2 months.

- ☒ Use Cost Explorer's granular filtering feature to perform an in-depth analysis of EC2 costs based on instance types.

Cost Explorer allows for in-depth cost analysis with granular filtering, enabling the architect to identify the root cause of unwanted scaling over the past two months. This is a built-in tool specifically designed for cost analysis. While granular filtering for hourly cost is for 14 days, it can be used monthly to show 2 months.

- ☐ Use graphs from the AWS Billing and Cost Management dashboard to compare EC2 costs based on instance types for the last 2 months.

The AWS Billing and Cost Management dashboard offers a high-level overview and is not ideal for the in-depth analysis required to identify the root cause of the vertical scaling.

- ☐ Use AWS Cost and Usage Reports to create a report and send it to an Amazon S3 bucket. Use Amazon QuickSight with Amazon S3 as a source to generate an interactive graph based on instance types.

While this approach can provide the required insights, it involves significant operational overhead due to the need to configure and manage AWS Cost and Usage Reports, Amazon S3, and Amazon QuickSight.

Overall Explanation:

Cost Explorer offers the best balance between providing the required in-depth analysis and minimizing operational overhead. It allows for filtering by instance type and time period, facilitating the identification of the root cause of the vertical scaling. Using Cost Explorer offers the least operational overhead than the other options.

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Question 48

Not Attempted

Networking



A company has migrated its data warehouse solution to AWS and utilizes an AWS Direct Connect connection. Corporate office users query the data warehouse using a visualization tool. The average size of a query returned by the data warehouse is 50 MB, while each webpage sent by the visualization tool is approximately 500 KB. Result sets returned by the data warehouse are not cached. Which solution will provide the LOWEST data transfer egress cost for the company?

- ☐ Host the visualization tool on premises and query the data warehouse directly over the internet.

Hosting the visualization tool on-premises and querying over the internet incurs data transfer costs for every query result that leaves AWS.

- ☐ Host the visualization tool in the same AWS Region as the data warehouse. Access it over the internet.

While hosting in the same region is better, accessing over the internet still incurs egress data transfer costs for the 50MB query results.

- ☐ Host the visualization tool on premises and query the data warehouse directly over a Direct Connect connection at a location in the same AWS Region.

Hosting the visualization tool on premises still requires the larger 50MB data transfer across the Direct Connect, representing a higher egress cost compared to option D.

- ☒ Host the visualization tool in the same AWS Region as the data warehouse and access it over a Direct Connect connection at a location in the same Region.

Hosting both the visualization tool and data warehouse in the same region, with access via Direct Connect minimizes egress costs. Only the smaller 500KB webpage data transfers out of AWS, leveraging the free data transfer within the same AWS region.

Overall Explanation:

Hosting the visualization tool in the same AWS Region as the data warehouse and accessing it over a Direct Connect connection within that Region provides the lowest data transfer egress cost. This setup minimizes the amount of data leaving the AWS environment, as the large 50 MB query results are processed within AWS, and only the smaller 500 KB webpages are transferred out via Direct Connect. Options A and B involve significant data transfer costs over the internet, while option C involves transferring the large 50 MB query results over Direct Connect.

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Question 49

Not Attempted

Databases



An online education platform is migrating its student records database to AWS. The company requires that its data remains consistently available and accessible across multiple AWS Regions with minimal administrative burden. The current PostgreSQL database must be migrated with a solution that guarantees uptime and disaster recovery capabilities across regions. Which of the following options offers the best solution?

- ☐ **✗** Migrate the PostgreSQL database to a PostgreSQL cluster on Amazon EC2 instances.
Managing a PostgreSQL cluster on EC2 requires significant operational overhead, including patching, backups, and failover configuration. This option does not provide the least amount of operational overhead.
- ☐ **✗** Migrate the PostgreSQL database to an Amazon RDS for PostgreSQL DB instance with the Multi-AZ feature enabled.
The Multi-AZ feature in RDS provides high availability within a single AWS Region, not across multiple Regions. This option does not meet the requirement of regional availability.
- ☒ **✓** Migrate the PostgreSQL database to an Amazon RDS for PostgreSQL DB instance. Create a read replica in another Region.
Creating a read replica in another region provides data availability and online access across multiple AWS Regions. Promoting the read replica offers DR capabilities and this is a managed service with low operational overhead.
- ☐ **✗** Migrate the PostgreSQL database to an Amazon RDS for PostgreSQL DB instance. Configure DB snapshots to be copied to another Region.

Using DB snapshots to be copied to another Region does not provide constant online availability. Snapshots require restoration to a new instance, causing downtime. This does not meet the requirement of the data to be 'available and online across multiple AWS Regions at all times'.

Overall Explanation:

Migrating the PostgreSQL database to an Amazon RDS for PostgreSQL DB instance and creating a read replica in another region is the most suitable solution. RDS read replicas can be created across regions, providing both data availability and DR capabilities with minimal operational overhead. The read replica can be promoted to be the main database in case of an outage in the primary region. Multi-AZ feature is for high availability inside a single region only. Snapshots are not online and requires restoration.

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Question 50

Not Attempted

Networking



A company hosts its web application on AWS using seven Amazon EC2 instances. The company requires that the IP addresses of all *healthy* EC2 instances be returned in response to DNS queries, with the possibility of returning up to eight. Which Route 53 routing policy should be used to meet this requirement?

☐ Simple routing policy

A Simple routing policy returns a single IP address. It does not support health checks to determine which IPs are healthy.

☐ Latency routing policy

Latency routing policy routes traffic based on the lowest latency to the resource and does not return all healthy IP addresses.

☒ Multivalue routing policy

Multivalue answer routing policy responds to DNS queries with up to eight healthy records selected at random. It also allows associating routing records with Route 53 health checks.

☐ Geolocation routing policy

Geolocation routing policy routes traffic based on the geographic location of the user and does not return all healthy IP addresses.

Overall Explanation:

The Multivalue routing policy allows Route 53 to respond to DNS queries with multiple healthy IP addresses for the same resource (up to eight), which is suitable in scenarios where multiple instances are serving the same purpose and need to be load balanced or failover capable. With the Multivalue routing policy, Route 53 returns multiple IP addresses in a random order to distribute the traffic across all healthy instances.

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Question 51

Not Attempted

Storage



A medical research lab produces data that is related to a new study. The lab wants to make the data available with minimum latency to clinics across the country for their on-premises, file-based applications. The data files are stored in an Amazon S3 bucket that has read-only permissions for each clinic. What should a solutions architect recommend to meet these requirements?

- ☒ Deploy an AWS Storage Gateway file gateway as a virtual machine (VM) on premises at each clinic

This is the correct answer. AWS Storage Gateway file gateway allows on-premises applications to access data stored in S3 buckets with low latency by caching frequently accessed data locally. It also maintains read-only permissions for the clinics, meeting the requirements.

- ☐ Migrate the files to each clinic's on-premises applications by using AWS DataSync for processing.

This is incorrect. While DataSync can transfer data, it involves migration rather than providing continuous, low-latency access to the S3 data. It doesn't directly address the requirement of making data available with minimum latency and maintaining read-only permissions.

- ☐ Deploy an AWS Storage Gateway volume gateway as a virtual machine (VM) on premises at each clinic.

This is incorrect. Volume gateway provides block-level access, which is not suitable for file-based applications. Also, it does not provide direct access to S3 objects like the file gateway does.

- ☐ Attach an Amazon Elastic File System (Amazon EFS) file system to each clinic's on-premises servers.

This is incorrect. Attaching an EFS file system to on-premises servers does not inherently provide low-latency access to data stored in an S3 bucket and introduces complexity.

Overall Explanation:

The best solution is to deploy an AWS Storage Gateway file gateway as a VM on premises at each clinic. This provides low-latency access to the data in S3 for file-based applications by caching frequently accessed data locally. It allows the clinics to directly access the data files stored in S3 with read-only permissions.

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Question 52

Not Attempted

Databases



A company hosts its e-commerce platform on a single EC2 instance, encompassing both the web server and database. To enhance resilience and cater to fluctuating user traffic, the company seeks a highly available and scalable architecture. What solution should a solutions architect propose to meet these requirements?

- ☐

Migrate the database to Amazon RDS with automatic backups enabled. Provision another EC2 instance manually within the same Availability Zone. Configure an Application Load Balancer in the same Availability Zone, targeting both instances.

This option doesn't provide adequate high availability because both EC2 instances reside in the same Availability Zone. A failure in that AZ would cause downtime. It also lacks automatic scaling capabilities.
- ☐

Migrate the database to Amazon Aurora with a read replica in the same Availability Zone as the existing EC2 instance. Manually launch another EC2 instance in the same Availability Zone. Configure an Application Load Balancer and set the two EC2 instances as targets.

While this option improves database performance and fault tolerance, it does not address the scalability requirements. The application instances are still fixed and not auto-scaled.
- ☒

Move the database to Amazon Aurora with a read replica in another Availability Zone. Create an Amazon Machine Image (AMI) from the EC2 instance. Configure an Application Load Balancer across two Availability Zones. Attach an Auto Scaling group that uses the AMI across two Availability Zones.

This approach provides both high availability and scalability. Aurora offers automatic failover and replication. The AMI facilitates easy EC2 instance replication across multiple AZs. The Application

Load Balancer and Auto Scaling group ensure even traffic distribution and automatic scaling based on demand.

- ⊗ Move the database to a separate EC2 instance and schedule backups to Amazon S3. Create an Amazon Machine Image (AMI) from the original EC2 instance. Configure an Application Load Balancer across two Availability Zones. Attach an Auto Scaling group that uses the AMI across two Availability Zones.

While this solution addresses scaling for the application tier, it leaves the database as a single point of failure, which is not highly available.

Overall Explanation:

The best solution is to move the database to Amazon Aurora with a read replica in another Availability Zone. Then, create an AMI from the EC2 instance and configure an Application Load Balancer in two Availability Zones, attaching an Auto Scaling group using the AMI. This provides high availability for both the application and the database layers and allows the website to scale to meet user demand.

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Question 53

Not Attempted

[Compute](#)

A company is launching an application on AWS. The application uses an Application Load Balancer (ALB) to direct traffic to at least two Amazon EC2 instances in a single target group. The instances are in an Auto Scaling group for each environment. The company requires a development environment and a production environment. The production environment will have periods of high traffic. Which solution will configure the development environment MOST cost-effectively?

- ⊗ Reconfigure the target group in the development environment to have only one EC2 instance as a target.
This option violates the requirement of having at least two EC2 instances in the target group.
- ⊗ Change the ALB balancing algorithm to least outstanding requests.
Changing the ALB balancing algorithm does not directly reduce the cost of the development environment.
- ⊗ Reduce the size of the EC2 instances in both environments.

Reducing the size of EC2 instances in the production environment might negatively impact its ability to handle periods of high traffic. It's best to keep the production environment's resources as required.

- ✔ Reduce the maximum number of EC2 instances in the development environment's Auto Scaling group.

Reducing the maximum number of EC2 instances in the development environment's Auto Scaling group limits the scaling capabilities and associated costs when scaling is not necessary. It still fulfills the minimum instance requirement.

Overall Explanation:

Reducing the maximum number of EC2 instances in the development environment's Auto Scaling group configures the development environment in the most cost-effective way. Development environments generally don't need the same level of scaling as production environments. This approach saves costs while maintaining a functional environment and satisfying the minimum instance requirement.

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Question 54

Not Attempted

Networking



A company hosts a web application on Amazon EC2 instances within private subnets across multiple Availability Zones. An internet-facing Application Load Balancer (ALB) has been implemented with the EC2 instances as the target group. Despite the configuration, internet traffic fails to reach the EC2 instances. What architectural adjustment is required to rectify this issue?

- ⊗ Replace the ALB with a Network Load Balancer. Configure a NAT gateway in a public subnet to allow internet traffic.

A NAT gateway is primarily for outbound internet access from private subnets. The issue is with inbound traffic not reaching the EC2 instances through the ALB.

- ⊗ Move the EC2 instances to public subnets. Add a rule to the EC2 instances' security groups to allow outbound traffic to 0.0.0.0/0.

Moving EC2 instances to public subnets is generally not recommended for security reasons. Outbound traffic rules are not relevant to the inbound traffic problem.

- ☐ Update the route tables for the EC2 instances' subnets to send 0.0.0.0/0 traffic through the internet gateway route. Add a rule to the EC2 instances' security groups to allow outbound traffic to 0.0.0.0/0.

This option focuses on outbound traffic via an internet gateway, which is not the primary issue. The problem lies with inbound traffic not reaching the EC2 instances.

- ☒ Create public subnets in each Availability Zone. Associate the public subnets with the ALB. Update the route tables for the public subnets with a route to the private subnets.

This allows the ALB to reside in public subnets, making it accessible from the internet. Updating the public subnet route tables to include routes to the private subnets is crucial for the ALB to forward traffic to the EC2 instances.

Overall Explanation:

The correct approach involves creating public subnets and associating them with the internet-facing ALB. This enables the ALB to receive internet traffic. Then, the route tables of the public subnets are updated to include routes to the private subnets, enabling the ALB to forward traffic to the EC2 instances residing in the private subnets. Note that subnets within a VPC can by default communicate with each other, therefore this option is not completely accurate. This question is poorly formulated.

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Question 55

Not Attempted

Databases



A company has deployed a database in Amazon RDS for MySQL. Due to increased transaction volume, the database support team is reporting slow reads against the DB instance and recommends adding a read replica. Which combination of actions should a solutions architect take *before* implementing this change? (Choose two.)

- ☐ Enable binlog replication on the RDS primary node.

Binary log replication is automatically enabled when creating a read replica in Amazon RDS. You don't need to enable it manually.

- ☐ Choose a failover priority for the source DB instance.

Failover priority is related to Multi-AZ configurations and automatic failover, but it is not specifically required when adding a read replica for scaling read operations.

- ✔ Allow long-running transactions to complete on the source DB instance.
Allowing long-running transactions to complete helps ensure data consistency during the creation of the read replica. It prevents the replica from being created in an inconsistent state.
- ✘ Create a global table and specify the AWS Regions where the table will be available.
Global tables are a feature of Amazon DynamoDB, not Amazon RDS for MySQL.
- ✔ Enable automatic backups on the source instance by setting the backup retention period to a value other than 0.
Automatic backups must be enabled on the source DB instance for read replicas to be created. RDS uses automated backups to create a snapshot of the primary DB instance to initialize the read replica.

Overall Explanation:

To create a read replica from an RDS for MySQL instance, automatic backups must be enabled on the source instance by setting the backup retention period to a value greater than 0. Also, it is best practice to allow any long-running transactions to complete on the source DB instance before initiating the read replica creation to avoid data inconsistencies.

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Question 56

Not Attempted

[Compute](#)

A company runs analytics software on Amazon EC2 instances. The software accepts job requests from users to process data that has been uploaded to Amazon S3. Users report that some submitted data is not being processed. Amazon CloudWatch reveals that the EC2 instances have a consistent CPU utilization at or near 100%. The company wants to improve system performance and scale the system based on user load. What should a solutions architect do to meet these requirements?

- ✘ Create a copy of the instance. Place all instances behind an Application Load Balancer.
While this option provides load balancing, it doesn't address the root cause of high CPU utilization or provide automatic scaling based on user load. It's also not the most efficient way to handle the job processing.
- ✘ Create an S3 VPC endpoint for Amazon S3. Update the software to reference the endpoint.

Creating an S3 VPC endpoint improves security and network performance, but it does not address the CPU utilization issue or provide a scaling mechanism based on user load.

- ⊗ Stop the EC2 instances. Modify the instance type to one with a more powerful CPU and more memory. Restart the instances.

This option addresses the CPU utilization issue, but it does not provide automatic scaling based on user load. Furthermore, it introduces downtime during the instance modification.

- ✓ Route incoming requests to Amazon Simple Queue Service (Amazon SQS). Configure an EC2 Auto Scaling group based on queue size. Update the software to read from the queue.

This option decouples the job requests from the processing instances using SQS. This allows to scale the number of instances based on the size of the queue, providing more resources when needed. Additionally, using an Auto Scaling group based on the queue size will automatically scale the number of instances up or down depending on the workload. Updating the software to read from the queue will allow it to process the job requests in a more efficient manner, improving the performance of the system.

Overall Explanation:

Routing incoming requests to Amazon SQS decouples the job requests from the EC2 instances, preventing overload. Configuring an EC2 Auto Scaling group based on queue size ensures that instances are added or removed automatically in response to user load, improving system performance and scalability. Updating the software to read from the queue ensures proper processing of the requests.

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Question 57

Not Attempted

Storage



A company is implementing a shared storage solution for a media application hosted in the AWS Cloud. The company requires the ability to use SMB clients to access data. The solution must be fully managed, minimizing administrative overhead. Which AWS solution meets these requirements?

- ⊗ Create an AWS Storage Gateway volume gateway. Create a file share that uses the required client protocol. Connect the application server to the file share.

AWS Storage Gateway volume gateway provides virtual disks using iSCSI, not SMB. It also requires management of the Storage Gateway appliance.

- ☐ Create an AWS Storage Gateway tape gateway. Configure tapes to use Amazon S3. Connect the application server to the tape gateway.
AWS Storage Gateway tape gateway provides virtual tapes via iSCSI, not SMB. It is designed for backup and archival purposes, not general file sharing.
- ☐ Create an Amazon EC2 Windows instance. Install and configure a Windows file share role on the instance. Connect the application server to the file share.
Creating an EC2 instance and configuring a Windows file share is not a fully managed solution. It requires manual administration of the operating system, file share configuration, and security.
- ☒ Create an Amazon FSx for Windows File Server file system. Attach the file system to the origin server. Connect the application server to the file system.
Amazon FSx for Windows File Server is a fully managed service that provides native support for Windows file system features and the SMB protocol. It simplifies file storage management in the AWS Cloud.

Overall Explanation:

Amazon FSx for Windows File Server directly supports SMB and is a fully managed service. The other options either don't support SMB natively or aren't fully managed. Therefore, Amazon FSx is the correct choice.

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Question 58

Not Attempted

Serverless



A company is designing an application that uses an AWS Lambda function to receive information through Amazon API Gateway and store it in an Amazon Aurora PostgreSQL database. During the proof-of-concept stage, the company had to significantly increase the Lambda quotas to handle the high data volumes. A solutions architect needs to recommend a new design to improve scalability and minimize configuration effort. Which solution meets these requirements?

- ☐ Refactor the Lambda function code to Apache Tomcat code that runs on Amazon EC2 instances. Connect the database by using native Java Database Connectivity (JDBC) drivers.
Refactoring requires significant effort and EC2 instances require management, making this option less desirable.

- ⊗ Change the platform from Aurora to Amazon DynamoDB. Provision a DynamoDB Accelerator (DAX) cluster. Use the DAX client SDK to point the existing DynamoDB API calls at the DAX cluster.

This requires a database migration from SQL to NoSQL, a substantial undertaking. DAX primarily benefits read operations, not the write-heavy workload described.

- ⊗ Set up two Lambda functions. Configure one function to receive the information. Configure the other function to load the information into the database. Integrate the Lambda functions by using Amazon Simple Notification Service (Amazon SNS).

SNS is a push mechanism that might overwhelm the second Lambda function, potentially recreating the initial bottleneck. It also doesn't guarantee message delivery in the same way SQS does.

- ✓ Set up two Lambda functions. Configure one function to receive the information. Configure the other function to load the information into the database. Integrate the Lambda functions by using an Amazon Simple Queue Service (Amazon SQS) queue.

Using SQS decouples the functions, allowing them to scale independently. The receiving function can quickly enqueue messages, and the loading function can process them at its own pace, improving scalability and handling high data volumes efficiently.

Overall Explanation:

The most scalable and efficient solution is to use two Lambda functions integrated via SQS. The first Lambda function receives the information and puts it into the SQS queue. The second Lambda function retrieves messages from the queue and loads them into the Aurora PostgreSQL database. This approach decouples the receiving and loading processes, allowing each to scale independently and handle high volumes of data. SQS provides asynchronous communication and buffering, which improves the overall system's resilience and scalability. Options A and B involve significant changes to the application architecture. Option C might create the same bottleneck as before.

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Question 59

Not Attempted

Networking



A company's security team requires network traffic to be captured using VPC Flow Logs. The logs need to be readily accessible for 90 days, after which they will be accessed only

intermittently. What configuration steps should a solutions architect take to meet these requirements cost-effectively?

- ☐ Use Amazon CloudWatch as the target. Configure the CloudWatch log group with a retention policy to expire logs after 90 days.
Using CloudWatch with a retention policy would delete the logs after 90 days, failing to meet the requirement of intermittent access after that period.
- ☐ Use Amazon Kinesis Data Streams as the target. Configure the Kinesis stream to retain logs for 90 days.
Kinesis Data Streams primarily focuses on real-time data ingestion and processing. While it can retain data for a period, it's not the most cost-effective solution for long-term storage and intermittent access, especially compared to S3.
- ☐ Use AWS CloudTrail as the target. Configure CloudTrail to save logs to an Amazon S3 bucket and enable S3 Intelligent-Tiering.
CloudTrail is designed for auditing API calls, not capturing network traffic data like VPC Flow Logs. Therefore, this is not the correct tool for the task.
- ☒ Use Amazon S3 as the target. Enable an S3 Lifecycle policy to transition the logs to S3 Standard-Infrequent Access (S3 Standard-IA) after 90 days.
Using S3 allows for long-term storage. Employing an S3 Lifecycle policy to transition logs to S3 Standard-IA after 90 days optimizes costs by moving less frequently accessed data to a cheaper storage tier while still allowing for intermittent access.

Overall Explanation:

The most cost-effective approach involves using Amazon S3 as the target for VPC Flow Logs. By enabling an S3 Lifecycle policy to transition the logs to S3 Standard-IA after the initial 90-day period, the company can optimize storage costs for data that is accessed intermittently while still maintaining availability. CloudWatch would delete the logs, Kinesis is not cost effective for storage, and CloudTrail is used for API call logging, not network traffic data.

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Question 60

Not Attempted

Networking



An Amazon EC2 instance is located in a private subnet within a new VPC. This subnet currently lacks outbound internet access, yet the EC2 instance requires the capability to download monthly security updates from an external vendor. What should a solutions architect implement to satisfy these conditions?

- ⊗ Create an internet gateway, and attach it to the VPC. Configure the private subnet route table to use the internet gateway as the default route.

This option provides direct internet access to the private subnet, which is not desired as the goal is to restrict unrestricted outbound internet access. By configuring the private subnet to use the internet gateway as the default route, it effectively makes the subnet public.

- ✔ Create a NAT gateway, and place it in a public subnet. Configure the private subnet route table to use the NAT gateway as the default route.

This approach allows the EC2 instance to access the internet through the NAT gateway, acting as a proxy. It provides controlled outbound internet access while maintaining the security of the private subnet.

- ⊗ Create a NAT instance, and place it in the same subnet where the EC2 instance is located. Configure the private subnet route table to use the NAT instance as the default route.

Placing the NAT instance in the private subnet where there is no internet access would prevent it from functioning. Additionally, NAT instances require more manual configuration and management.

- ⊗ Create an internet gateway, and attach it to the VPC. Create a NAT instance, and place it in the same subnet where the EC2 instance is located. Configure the private subnet route table to use the internet gateway as the default route.

This combines the use of an internet gateway and a NAT instance, which is unnecessary. It introduces complexity and adds a NAT instance that requires additional management. The NAT instance wouldn't be useful in this scenario because it's in a private subnet with no internet access.

Overall Explanation:

The correct solution is to create a NAT gateway in a public subnet and configure the private subnet's route table to route traffic through the NAT gateway. This allows outbound traffic to the internet for updates while keeping the EC2 instance in a private, secure subnet. NAT Gateways are the preferred AWS solution.

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Question 61

Not Attempted

Storage



A solutions architect is designing a system to store client case files. The files are critical company assets and their persistence is paramount. The number of files will grow significantly over time. The files must be simultaneously accessible from multiple application servers that run on Amazon EC2 instances. The solution must provide built-in redundancy. Which storage solution BEST meets these requirements?

☒ Amazon Elastic File System (Amazon EFS)

EFS is a scalable and fully managed file storage service that can be accessed concurrently from multiple EC2 instances. It offers built-in redundancy by storing data across multiple Availability Zones (AZs) within a region. EFS automatically grows and shrinks as you add and remove files, eliminating the need for manual management or provisioning.

☐ Amazon Elastic Block Store (Amazon EBS)

EBS is a block-level storage service typically used for attaching to individual EC2 instances. It does NOT provide concurrent access to multiple instances, making it unsuitable for this scenario.

☐ Amazon S3 Glacier Deep Archive

S3 Glacier Deep Archive is a long-term archival storage service and may not be suitable for active file access and simultaneous access from multiple application servers due to its retrieval times.

☐ AWS Backup

AWS Backup is a centralized backup management service and does NOT provide the required simultaneous file access and redundancy features for a primary storage solution.

Overall Explanation:

Amazon EFS is the BEST solution because it provides a scalable, fully managed file system that can be accessed concurrently by multiple EC2 instances. It also offers built-in redundancy by storing data across multiple AZs. EBS is block storage for a single instance. Glacier is for archiving, not active use. AWS Backup is for backups, not the primary storage location.

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Question 62

Not Attempted

Security



A solutions architect has created two IAM policies: Policy1 and Policy2. Both policies are attached to an IAM group. A cloud engineer is added as an IAM user to the IAM group. Which action will the cloud engineer be able to perform?

Policy 1

```
{
  "Version": "2012-10-17",  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "iam:Get*",
        "iam:List*",
        "kms:List*",
        "ec2:*",
        "ds:*",
        "logs:Get*",
        "logs:Describe*"
      ],
      "Resource": "*"
    }
  ]
}
```

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Policy 2

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Deny",
      "Action": "ds:Delete*",
      "Resource": "*"
    }
  ]
}
```

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⊗ Deleting IAM users

Policy1 only grants get and list permission on IAM users, so the user cannot delete them.

⊗ Deleting directories

Policy2 explicitly denies the delete-directory action, so the user cannot delete directories.

✓ Deleting Amazon EC2 instances

Policy1 grants full control of EC2 instances, allowing the user to delete them.

⊗ Deleting logs from Amazon CloudWatch Logs

Policy1 only grants get and describe permission on CloudWatch Logs, so the user cannot delete them.

Overall Explanation:

Policy1 grants full control of EC2 instances (ec2:*), which includes the ability to delete them. Policy2 explicitly denies the deletion of directories. Policy1 only grants get and list permissions on IAM users and get and describe permissions on CloudWatch logs.

Therefore, the cloud engineer will only be able to delete Amazon EC2 instances.

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Question 63

Not Attempted

Security



A company recently migrated a three-tier application to a VPC. During a security review, it's discovered that the principle of least privilege is not being enforced in the Amazon EC2 security group ingress and egress rules between the application tiers. Which of the following actions should a solutions architect take to remediate this security issue?

- ☐ Create security group rules using the instance ID as the source or destination.
Using instance IDs is not the best approach as it ties security rules to specific instances rather than the application tier, making management more complex and less scalable.
- ☒ Create security group rules using the security group ID as the source or destination.
Using security group IDs allows precise control of traffic between application tiers, enforcing the principle of least privilege by only allowing necessary communication.
- ☐ Create security group rules using the VPC CIDR blocks as the source or destination.
Using VPC CIDR blocks is too broad and does not adhere to the principle of least privilege, as it allows traffic from any IP address within the VPC.
- ☐ Create security group rules using the subnet CIDR blocks as the source or destination.
Using subnet CIDR blocks is also too broad, allowing traffic from any IP address within the subnet and not adhering to the principle of least privilege.

Overall Explanation:

The best way to enforce the principle of least privilege is to use security group IDs as the source or destination in the security group rules. This allows you to control traffic between application tiers precisely, allowing only the necessary communication. Options using instance IDs, VPC CIDR blocks, or subnet CIDR blocks are too broad and do not provide the granular control needed to minimize access.

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Question 64

Not Attempted

Databases



A company's e-commerce checkout process writes order data to a database and then triggers a payment processing service. Users are encountering timeouts during checkout, and when they resubmit the form, duplicate orders are created for the same transaction. To prevent multiple order creation, how should a solutions architect redesign this workflow?

- ☐ ⓧ Configure the web application to send order details to Amazon Kinesis Data Firehose. Have the payment service retrieve and process messages from Kinesis Data Firehose.
Kinesis Data Firehose is primarily for streaming data to data lakes or analytics services, not for guaranteed ordered processing for transactional workflows. It doesn't prevent duplicate order processing in case of retries.
- ☐ ⓧ Create an AWS CloudTrail rule to invoke an AWS Lambda function based on application path requests. Use Lambda to query the database, initiate payment processing, and pass the order information.
CloudTrail is for auditing API calls, not for building transactional workflows. Using Lambda triggered by CloudTrail is not suitable for reliable order processing and doesn't prevent duplicate orders.
- ☐ ⓧ Store the order in the database. Send a message containing the order number to Amazon Simple Notification Service (Amazon SNS). Configure the payment service to poll Amazon SNS, retrieve messages, and process the order.
Amazon SNS is a publish/subscribe service that does not guarantee message delivery or order. Polling SNS for messages is inefficient and doesn't prevent multiple orders from being processed.
- ☒ ✓ Store the order in the database. Send a message containing the order number to an Amazon Simple Queue Service (Amazon SQS) FIFO queue. Configure the payment service to retrieve the message, process the order, and then delete the message from the queue.
This approach utilizes a FIFO queue to ensure that orders are processed sequentially and exactly once. By deleting the message after successful processing, duplicate processing is prevented. The queue decouples the order creation from payment processing allowing for retries without creating duplicate orders.

Overall Explanation:

Using an Amazon SQS FIFO queue ensures that each order is processed exactly once and in the order it was received. Storing the order in the database first allows for retries without order loss. Sending a message to the FIFO queue ensures that the payment service processes orders sequentially, and deleting the message after processing prevents

duplicates in case of retries. This approach effectively addresses the issue of multiple orders being created during timeouts.

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Question 65

Not Attempted

Storage



A solutions architect is designing a document review application that utilizes an Amazon S3 bucket for storing documents. The application must ensure that documents cannot be accidentally deleted and that all previous versions of the documents are recoverable. Users need the ability to download, modify, and re-upload documents seamlessly. Which combination of actions should the architect implement to meet these requirements? (Choose two.)

☐ Enable a read-only bucket ACL.

A read-only bucket ACL would restrict users from modifying or uploading documents, which contradicts the requirement for users to be able to download, modify, and upload documents.

☒ Enable versioning on the bucket.

Enabling versioning allows multiple versions of objects in the S3 bucket to be stored. This ensures that all versions of the documents are available, even if they are accidentally overwritten or deleted.

☐ Attach an IAM policy to the bucket.

While an IAM policy can control access permissions to the bucket, it doesn't specifically address the requirements of preventing accidental deletion or ensuring the availability of all document versions.

☒ Enable MFA Delete on the bucket.

Enabling MFA Delete adds an extra layer of protection against accidental or unauthorized deletion of objects in the bucket. This requires multi-factor authentication to delete objects, preventing accidental deletions.

☐ Encrypt the bucket using AWS KMS.

Encryption primarily focuses on data protection and security rather than versioning or deletion prevention. While important for security, it doesn't directly address the stated requirements.

Overall Explanation:

To prevent accidental deletion of documents and ensure all versions are available, the architect should enable versioning on the S3 bucket and enable MFA Delete. Versioning keeps a history of all changes, while MFA Delete requires multi-factor authentication for deletions, preventing accidental or unauthorized removal of objects.

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